

Urinary Tract Obstruction

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KEY POINTS

- Hydronephrosis and obstructive uropathy are not interchangeable terms—dilation of the renal pelvis and calices can occur without obstruction, and urinary tract obstruction may occur in the absence of hydronephrosis.
- Urinary tract obstruction may be congenital or acquired. Acquired causes can be intrinsic or extrinsic.
- There is no single diagnostic technique that can safely diagnose obstruction. Therefore obstruction must still be considered in patients with worsening renal function, chronic azotemia, or acute changes in renal function or urine output, even in the absence of hydronephrosis.
- Both intrarenal and extrarenal factors combine to decrease glomerular filtration rate and renal blood flow during and immediately after release of obstruction.
- Urinary tract obstruction disrupts the ability to concentrate and dilute urine and impairs epithelial sodium, proton, and bicarbonate transport due to downregulation of most transporter proteins in the tubules.
- Urinary tract obstruction results in development of tubulointerstitial fibrosis with irreversible kidney injury due to a complex series of mechanisms involving multiple hormonal systems as well as numerous cellular and molecular pathways.

In adults, 1.5 to 2.0 L of urine flows daily from the renal papillae through the ureter, bladder, and urethra in an uninterrupted, unidirectional flow. Any obstruction of urinary flow at any point along the urinary tract may cause retention of urine and increased retrograde hydrostatic pressure, leading to kidney damage and interference with waste and water excretion, as well as fluid and electrolyte homeostasis. Because the extent of recovery of renal function in obstructive nephropathy is related inversely to the extent and duration of obstruction, prompt diagnosis and relief of obstruction are essential for effective management. Fortunately, urinary tract obstruction in most cases is a highly treatable form of kidney disease.

Several terms describe urinary tract obstruction, and definitions may vary.^{1–3} In the following discussion “hydronephrosis” is defined as a dilation of the renal pelvis and calices proximal to the point of obstruction. “Obstructive uropathy” refers to blockage of urine flow due to a functional or structural derangement anywhere from the tip of the urethra back to the renal pelvis that increases pressure proximal to the site of obstruction. Obstructive uropathy may or may not result in renal parenchymal damage. Such functional or pathologic parenchymal damage is referred to as *obstructive nephropathy*. It should be noted that hydronephrosis and obstructive uropathy are not interchangeable terms—dilation of the renal pelvis and calices can occur without obstruction, and

urinary tract obstruction may occur in the absence of hydronephrosis.

PREVALENCE AND INCIDENCE

The incidence of urinary tract obstruction varies widely among different populations and depends on concurrent medical conditions, sex, and age. Unfortunately, epidemiological reports have been based on the studies of selected “populations,” such as women with high-risk pregnancies and data from autopsy series. In the United States it has been estimated that 166 patients per 100,000 population had a presumptive diagnosis of obstructive uropathy on admission to hospitals in 1985.⁴ The introduction of routine prenatal ultrasound (US) scanning resulted in an increasing number of infants suspected with urinary tract obstruction,⁵ and with the increasing age of the population during the past 25 years the incidence of obstructive uropathy may be expected to increase even more.

A review of 59,064 autopsies of subjects varying in age from neonate to 80 years noted hydronephrosis as a finding in 3.1% of cases (3.3% in males and 2.9% in females). In subjects under age 10 years, representing 1.5% of all autopsies, the principal causes of urinary tract obstruction were ureteral or urethral strictures, or neurologic abnormalities. It is unclear how frequently these abnormalities represented incidental findings, as opposed to being recognized clinically. Until the age of 20, there was no substantial sex difference in frequency of abnormalities (for details, please also see Chapter 72). Between the ages of 20 and 60 years, urinary tract obstruction was more frequent among women than among men, mainly due to the effects of uterine cancer and pregnancy. Above the age of 60 years, prostatic disease raised the frequency of urinary tract obstruction among men above that observed among women.

In children under age 15 years, obstruction occurred in 2% of autopsies. Hydronephrosis was found in 2.2% of the boys and 1.5% of the girls; 80% of the hydronephrosis that did occur was found in subjects aged under 1 year.⁶ Consistent with this, another autopsy series of 3172 children identified urinary tract abnormalities in 2.5%. Hydroureter and hydronephrosis were the most common findings, representing 35.9% of all cases.⁷ However, it was not clear what proportion of cases was diagnosed clinically before death.

Because a high proportion of these autopsy-detected cases of obstruction likely went undetected during life, the overall prevalence of urinary tract obstruction is very likely far greater than reports suggest. This conclusion is reinforced by the fact that there are several common but temporary causes of obstruction, such as pregnancy and renal calculi.

CLASSIFICATION

Classification of urinary tract obstruction can be by duration (i.e., acute or chronic⁸), by whether it is congenital or acquired, and by its location (upper or lower urinary tract, supravescical, vesical, or subvesical, etc.). Acute obstruction may be associated with sudden onset of symptoms. Upper urinary tract [ureter or ureteropelvic junction (UPJ)] obstruction may present with renal colic. Lower tract (bladder or

urethra) obstruction may present with disorders of micturition. By contrast, chronic urinary tract obstruction may develop insidiously and present with few or only minor symptoms, and with more general manifestations. For example, recurrent urinary tract infections, bladder calculi, and progressive renal insufficiency may all result from chronic obstruction. Congenital causes of obstruction arise from developmental abnormalities, whereas acquired lesions develop after birth, as a result of either disease processes or medical interventions.⁹

ETIOLOGY

Because congenital and acquired urinary tract obstructions differ to a great degree in cause and clinical course, they will be described separately.

CONGENITAL CAUSES OF OBSTRUCTION

Congenital anomalies may obstruct the urinary tract at any level from the UPJ to the tip of urethra, and the obstruction may damage one or both kidneys (Box 37.1). Although some lesions occur rarely, as a group they represent an important cause of urinary tract obstruction because in younger patients they often lead to severe renal impairment and may result in catastrophic end-stage renal disease.¹⁰ Thus this condition is also presented in detail in Chapter 72.

The widespread use of fetal ultrasonography, and its increasing sensitivity, has led to early detection in an increasing number of cases. With an estimated prevalence of 2% to 5.5%, dilatation of the renal collecting system is the most common ultrasonographic abnormality found in the fetal

Box 37.1 Congenital Causes of Urinary Tract Obstruction

Ureteropelvic Junction

- Ureteropelvic junction obstruction
- Proximal and middle ureter
- Ureteral folds
- Ureteral valves
- Strictures
- Benign fibroepithelial polyps
- Retrocaval ureter

Distal Ureter

- Ureterovesical junction obstruction
- Vesicoureteral reflux
- Prune-belly syndrome
- Ureterocele

Bladder

- Bladder diverticula
- Neurologic conditions (e.g., spina bifida)

Urethra

- Posterior urethral valves
- Urethral diverticula
- Anterior urethral valves
- Urethral atresia
- Labial fusion

urinary tract.¹¹ In cases of severe obstruction early detection may lead to termination of the pregnancy or attempts to ameliorate the obstruction in utero.^{10,12,13} However, US may detect mild obstruction of unknown clinical significance.^{10,12} In brief, UPJ obstruction is the most common cause of hydronephrosis in fetuses¹⁴ and young children,¹⁵ with a reported incidence of 5 cases per 100,000 population per year,¹⁶ and it may affect adults as well.¹⁷ There is considerable controversy as to whether all cases of obstruction early in life are clinically significant. The widespread use of fetal US has resulted in detection of many cases that remain asymptomatic and may resolve spontaneously with simple follow-up of the child.^{3,18} Despite the numerous cases, there is a lack of consistency with respect to the nomenclature and grading systems used in the clinical risk assessment of infants with antenatal hydronephrosis.⁵ Although most cases of congenital UPJ obstruction are diagnosed prenatally by US,¹⁹ the most common neonatal clinical presentation is a flank or abdominal mass.²⁰ By contrast, adults generally present with flank pain.¹⁷ Because intermittent obstruction may produce symptoms that mimic those of gastrointestinal disease, diagnosis may be delayed. At any age, UPJ obstruction may be associated with kidney stones, hematuria, hypertension, or recurrent urinary tract infection.^{16,17} A detailed presentation of the different causes of UPJ obstruction is provided in Chapter 72, where a thorough presentation of the pathophysiology of proximal and distal congenital ureter obstruction is discussed.

Congenital bladder outlet obstruction may be caused by mechanical or functional factors and will also be discussed in Chapter 72.

Because operative complications may be high,²¹ the use of fetal^{13,22} or neonatal^{22,23} surgery for the relief of obstruction remains controversial.^{10,12,13} Although bilateral obstruction requires intervention, patients with unilateral hydronephrosis are often followed without surgery, but with aggressive observation to identify the approximately 20% of patients with congenital hydronephrosis who require pyeloplasty.^{23,24} Indications for surgery in unilateral hydronephrosis include symptoms of obstruction or impaired function in a presumably salvageable hydronephrotic kidney.

ACQUIRED CAUSES OF OBSTRUCTION

INTRINSIC CAUSES

Acquired causes of obstruction may be intrinsic to the urinary tract (i.e., resulting from intraluminal or intramural processes) or may arise from causes extrinsic to it (Box 37.2). Intrinsic causes of obstruction may be considered according to anatomic location.

Intrinsic intraluminal causes of obstruction may be intrarenal or extrarenal. Intrarenal causes arise from formation of casts or crystals within the renal tubules. These include uric acid nephropathy²⁵; deposition of crystals of drugs that precipitate in the urine, including sulfonamides,²⁶ acyclovir,²⁷ indinavir,²⁸ and ciprofloxacin²⁹; and multiple myeloma.³⁰ Uric acid nephropathy usually results from the large uric acid load released when alkylating agents abruptly kill large numbers of tumor cells in the treatment of patients with malignant hematopoietic neoplasms. The risk of uric acid nephropathy relates directly to plasma uric acid concentrations.²⁵ Uric acid nephropathy may also occur in the setting of disseminated adenomatous carcinoma of the gastrointestinal

tract.³¹ Sulfonamide crystal deposition, once a common occurrence, became rare with the introduction of sulfonamides that are more soluble in acid urine than earlier drugs were. Sulfadiazine has been used as antiretroviral therapy because it is relatively lipophilic and penetrates the brain well, making it an excellent treatment for toxoplasmosis in patients with acquired immunodeficiency syndrome (AIDS). However, the same lipophilicity makes the drug prone to the formation of intrarenal crystals, which can lead to acute kidney injury when the drug is given in large doses.^{26,32} Ciprofloxacin may also precipitate in the tubular fluid, resulting in crystalluria with stone formation and urinary tract obstruction.²⁹ A common renal complication of multiple myeloma is “myeloma kidney,” a condition also known as myeloma cast nephropathy. The renal lesions (casts) are directly related to the production of monoclonal immunoglobulin free light chains (FLCs), which coprecipitate with Tamm–Horsfall glycoprotein (THP) in the lumen of the distal nephron, obstructing tubular fluid flow.^{33,34} Promising experiments have identified the determinants of the molecular interaction between FLCs and THP, which permitted development of a peptide that demonstrated strong inhibitory capability in the binding of FLCs to THP in vitro.³⁴

Several intrinsic intraluminal, extrarenal, or intraureteral processes may also cause obstruction. Nephrolithiasis represents the most common cause of ureteral obstruction in younger men.³⁵ In the US population the prevalence of symptomatic kidney stone of adults aged 20 to 74 years was estimated from self-reported incidents between 1988 and 1994 to afflict 5.2% of adults (6.3% males and 4.1% females).³⁶ The significance of this number is also reflected by the large number of hospital admissions due to calculus of the kidney and ureters, amounting to 166,000 hospital stays in 2006.³⁷ Calcium oxalate stones occur most commonly. Obstruction caused by such stones occurs sporadically and tends to be acute and unilateral, and usually without a long-term impact on renal function. Of course, when a stone obstructs a solitary kidney the result can be anuric or oliguric acute kidney injury. Less common types of stones, such as struvite (ammonium–magnesium–sulfate) and cysteine stones, more frequently cause significant renal damage because these substances accumulate over time, and often form staghorn calculi. Stones tend to lodge and to obstruct urine flow at narrowings along the ureter, including the UPJ, the pelvic brim (where the ureter arches over the iliac vessels), and the ureterovesical junction.

Other processes that cause ureteral obstruction include papillary necrosis, blood clots, and cystic inflammation. Papillary necrosis³⁸ may result from sickle cell disease or trait, amyloidosis,³⁹ analgesic abuse, acute pyelonephritis, or diabetes mellitus. Renal allografts may develop papillary necrosis as well.⁴⁰ Acute obstruction may even require surgical intervention.⁴¹ Blood clots secondary to a benign or malignant lesion of the urinary tract or cystic inflammation of the ureter (ureteritis cystica) can also lead to obstruction and hydronephrosis.⁴²

Intrinsic intramural processes that cause obstruction include failure of micturition or more rarely of ureteral peristalsis. Bladder storage of urine and micturition require complex interplay of spinal reflexes, midbrain, and cortical function.⁴³ Neurologic dysfunction⁴⁴ occurring in diabetes mellitus, multiple sclerosis, spinal cord injury, cerebrovascular disease,

Box 37.2 Acquired Causes of Urinary Tract Obstruction**Intrinsic Processes****Intraluminal**

Intrarenal
 Uric acid nephropathy
 Sulfonamides
 Acyclovir
 Indinavir
 Multiple myeloma
 Intraureteral
 Nephrolithiasis
 Papillary necrosis
 Blood clots
 Fungus balls

Intramural

Functional
 Diseases
 Diabetes mellitus
 Multiple sclerosis
 Cerebrovascular disease
 Spinal cord injury
 Parkinson disease
 Drugs
 Anticholinergic agents
 Levodopa (α -adrenergic properties)
 Anatomic
 Ureteral strictures
 Schistosomiasis
 Tuberculosis
 Drugs (e.g., nonsteroidal antiinflammatory agents)
 Ureteral instrumentation
 Urethral strictures
 Benign or malignant tumors of the renal pelvis, ureter, bladder

Extrinsic Processes**Reproductive Tract**

Females
 Uterus
 Pregnancy
 Tumor (fibroids, endometrial or cervical cancer)
 Endometriosis
 Uterine prolapse
 Ureteral ligation (surgical)
 Ovary
 Tubo-ovarian abscess
 Tumor
 Cyst

Males
 Benign prostatic hyperplasia
 Prostate cancer

Malignant Neoplasms

Genitourinary Tract
 Tumors of kidney, ureter, bladder, urethra
 Other sites
 Metastatic spread
 Direct extension

Gastrointestinal System

Crohn's disease
 Appendicitis
 Diverticulosis
 Chronic pancreatitis with pseudocyst formation
 Acute pancreatitis

Vascular System

Arterial Aneurysms
 Abdominal aortic aneurysm
 Iliac artery aneurysm
 Venous
 Ovarian vein thrombophlebitis
 Vasculitides
 Systemic lupus erythematosus
 Polyarteritis nodosa
 Wegener granulomatosis
 Henoch-Schönlein purpura

Retroperitoneal Processes

Fibrosis
 Idiopathic
 Drug induced
 Inflammatory
 Ascending lymphangitis of the lower extremities
 Chronic urinary tract infection
 Tuberculosis
 Sarcoidosis
 Iatrogenic (multiple abdominal surgical procedures)
 Enlarged retroperitoneal nodes
 Tumor invasion
 Tumor mass
 Hemorrhage
 Urinoma

Biologic Agents

Actinomycosis

and Parkinson disease can result from upper motor neuron damage. These can produce a variety of forms of bladder dysfunction. If the bladder fails to empty properly, it can remain filled most of the time, resulting in chronic increased intravesical pressure, which is transmitted retrograde into the ureters and to the renal pelvis and kidney. In addition, failure of coordination of bladder contraction with the opening of the urethral sphincter may lead to bladder hypertrophy. In this setting, bladder filling requires increased hydrostatic pressures to stretch the hypertrophic detrusor muscle. Again the increased pressure in the bladder is transmitted up the urinary tract to the ureters and renal pelvis. Lower spinal tract injury may result in a flaccid, atonic

bladder and failure of micturition, as well as recurrent urinary tract infections.

Various drugs may cause intrinsic intramural obstruction by disrupting the normal function of the smooth muscle of the urinary tract. Anticholinergic agents⁴⁵ may interfere with bladder contraction, whereas levodopa⁴⁶ may mediate an α -adrenergic increase in urethral sphincter tone, resulting in increased bladder outlet resistance. Chronic use of tiaprofenic acid (Surgam) can cause severe cystitis with subsequent ureteral obstruction.⁴⁷ In all circumstances when the bladder does not void normally, renal damage may develop as a consequence of recurrent urinary tract infections and back pressure produced by the accumulation of residual urine.

Acquired anatomic abnormalities of the wall of the urinary tract include ureteral strictures and benign as well as malignant tumors of the urethra, bladder, ureter, or renal pelvis.⁴⁸ Ureteral strictures may result from radiation therapy in children⁴⁹ and in adults⁵⁰ treated for pelvic or lower abdominal cancers, such as cervical cancer, or nowadays rarely as a result of analgesic abuse.⁵¹ Strictures may also develop as a complication of ureteral instrumentation or surgery.

Infectious organisms may also produce intrinsic obstruction of the urinary tract. *Schistosoma haematobium* afflicts nearly 100 million people worldwide. Although active infection can be treated and obstructive uropathy may resolve, chronic schistosomiasis (bilharziasis) may develop in untreated cases, leading to irreversible ureteral or bladder fibrosis and obstruction.⁵² Of other infections, the incidence of genitourinary tuberculosis has remained constant over the 30 years, amounting to 3% to 5% of patients with tuberculosis.⁵³ Mycoses caused by *Candida albicans* or *Candida tropicalis* may also result in obstruction due to intraluminal obstruction (fungus ball) or invasion of the ureteral wall.⁵⁴

EXTRINSIC CAUSES

Acquired extrinsic urinary tract obstruction occurs in a wide variety of settings. The relatively high frequency of obstructive uropathy from processes in the female reproductive tract such as pregnancy and pelvic neoplasms results in higher rates of urinary tract obstruction in younger women than in younger men.² The advent of routine abdominal and fetal ultrasonography in pregnant women has revealed that more than two-thirds of women entering their third trimester demonstrate some degree of dilation of the collecting system,⁵⁵ most often resulting from mechanical ureteral obstruction.⁵⁵ This temporary form of obstruction is usually observed above the point at which the ureter crosses the pelvic brim, and affects the right ureter more often than the left.⁵⁵ The vast majority of these cases are subclinical and appear to resolve completely soon after delivery.⁵⁶ Clinically significant obstructive uropathy in pregnancy almost always presents with flank pain.⁵⁷ In these cases, ultrasonography serves as a useful initial screening test,²⁰ and magnetic resonance imaging (MRI) can be used if the US is not conclusive.⁵⁷ Of course, the diagnostic evaluation must be tailored to minimize fetal radiation exposure. If the obstruction is significant, a ureteral stent can be placed cystoscopically, and its efficacy can be monitored with repeated follow-up ultrasonography.⁵⁸ The stent can be left in place for the duration of pregnancy, if needed. Clinically significant ureteral obstruction is rare in pregnancy, and bilateral obstruction leading to acute kidney injury is exceptionally rare.⁵⁷ Conditions in pregnancy that may predispose to obstructive uropathy and acute kidney injury include multiple fetuses, polyhydramnios, an incarcerated gravid uterus, or a solitary kidney.⁵⁶

Pelvic malignancies, especially cervical adenocarcinomas, represent the second most common cause of extrinsic obstructive uropathy in women.⁵⁹ In older women, uterine prolapse and other failures of normal pelvic floor tone may cause obstruction, with hydronephrosis developing in 5% of patients.⁶⁰ In this setting prolapse may lead to compression of the ureter by uterine blood vessels. In addition, prolapse has been associated with urinary tract infection, sepsis, pyonephrosis, and renal insufficiency. Prolapse of other pelvic organs due to weakening of the pelvic floor may also result

in obstruction.⁶⁰ Various benign pelvic abnormalities may cause ureter obstruction, including uterine tumors or cystic ovary and pelvic inflammatory disease, particularly a tubo-ovarian abscess. Pelvic lipomatosis, a disease with an unclear etiology seen more often in men, is another rare reason for compressive urinary tract obstruction.⁶¹

Although endometriosis only rarely results in ureteral obstruction,⁶² it should be included in the differential diagnosis any time a premenopausal woman presents with unilateral obstruction. The onset of obstruction may be insidious, and the process is usually confined to the pelvic portion of the ureter.⁶² Ureteral involvement may be intrinsic or extrinsic, with extrinsic compression arising principally from adhesions associated with the endometriosis. Because ureteral involvement may come on slowly and may be unilateral, it is important to screen for obstructive uropathy in advanced cases of endometriosis.⁶² In this case using computed tomography (CT) is preferred, because US may not reveal hydronephrosis if adhesions are preventing dilatation of the ureter above the site of obstruction.⁶³ When surgery of any kind is contemplated in patients with endometriosis, it is all the more important to image the ureters, because they cross the anticipated surgical field and may well be near, or attached to, adhesions.^{62,63} Note that 52% of inadvertent ligations of the ureter in abdominal and retroperitoneal operations occur in gynecologic procedures.⁶⁴

Above the age of 60 years, obstructive uropathy occurs more commonly in men than in women. Benign prostatic hyperplasia, which is by far the most common cause of urinary tract obstruction in men, produces some symptoms of bladder outlet obstruction in 75% of men aged 50 years and older.^{65,66} It is likely that the proportion of affected older men would be higher if physicians routinely took a detailed history for symptoms.^{65,66} Presenting symptoms of bladder outlet obstruction include difficulty initiating micturition, weakened urinary stream, dribbling at the end of micturition, incomplete bladder emptying, and nocturia. The diagnosis may be established by history and urodynamic studies, as well as imaging in some cases.^{65–67}

Malignant genitourinary tumors occasionally cause urinary tract obstruction. Bladder cancer is the second most common cause (after cervical cancer) of malignant obstruction of the ureter.² Despite stage migration to more organ-confined disease in the era of prostate-specific antigen, obstruction due to prostate cancer compressing the bladder neck and invading the ureteral orifices is still relatively common.⁶⁸ Urinary tract obstruction in advanced and metastatic prostate cancer can have a varied presentation because it may occur in multiple anatomic locations including the ureter and pelvic lymph nodes.⁶⁹ Although urothelial tumors of the renal pelvis, ureter, and urethra are very rare, they also may lead to urinary obstruction.⁷⁰

Several gastrointestinal processes may rarely cause obstructive uropathy. Inflammation in Crohn's disease may extend into the retroperitoneum, leading to obstruction of the ureters,⁷¹ usually on the right side.⁷² In addition, several gastrointestinal diseases may cause oxalosis, leading to nephrolithiasis.⁷³ Appendicitis may lead to retroperitoneal scarring or abscess formation in children and young adults,⁷⁴ leading to obstruction of the right ureter. Diverticulitis in older patients⁷⁵ may rarely cause obstruction of the left ureter. Fecaloma is another rare cause of bilateral ureteral

obstruction.⁷⁶ Chronic pancreatitis with pseudocyst formation sometimes causes left ureteral obstruction,⁷⁷ and may very rarely cause bilateral obstruction.⁷⁸ Acute pancreatitis may result in right-sided obstruction.⁷⁹

Vascular abnormalities or diseases may also lead to obstruction. Abdominal aortic aneurysm is the most common vascular cause of urinary obstruction,⁸⁰ which may be caused by direct pressure of the aneurysm on the ureter or associated retroperitoneal fibrosis. Aneurysms of the iliac vessels may also cause obstruction of the ureters as they cross over the vessels.⁸⁰ Rarely, the ovarian venous system may cause right ureteral obstruction.⁸¹ In addition, and also rarely, vasculitis caused by systemic lupus erythematosus,⁸² polyarteritis nodosa,⁸³ Wegener granulomatosis,⁸⁴ and Henoch-Schönlein purpura^{85,86} has been reported to cause obstruction.

Retroperitoneal processes, such as tumor invasion leading to compression, as well as retroperitoneal fibrosis, can result in obstruction. The major extrinsic causes of retroperitoneal obstruction, accounting for 70% of all cases, are due to tumors of the colon, bladder, prostate, ovary, uterus, or cervix.^{2,87,88} When idiopathic, retroperitoneal fibrosis^{87,88} usually involves the middle third of the ureter and affects men and women equally, predominantly those in the 5th and 6th decades of life.⁸⁸ Retroperitoneal fibrosis may also be drug induced (e.g., methysergide), or it may occur as a consequence of scarring from multiple abdominal surgical procedures.⁸⁸ It may also be associated with conditions as varied as gonorrhea, sarcoidosis, chronic urinary tract infections, Henoch-Schönlein purpura, tuberculosis, biliary tract disease, and inflammatory processes of the lower extremities with ascending lymphangitis.⁸⁸

Malignant neoplasms can obstruct the urinary tract by direct extension or by metastasis,⁸⁹ which may be managed by retrograde stenting as a practical but guarded treatment and should be tailored to each patient.⁸⁹ As noted earlier, cervical cancer is the most common obstructing malignant neoplasm, followed by bladder cancer.^{2,90,91} Rare childhood tumors such as pelvic neurofibromas can induce upper urinary tract obstruction in up to 60% of patients.⁹² Wilms tumor may obstruct via local compression of the renal pelvis.⁹³ Miscellaneous inflammatory processes can also result in obstruction. These include granulomatous causes such as sarcoidosis⁹⁴ and chronic granulomatous disease of childhood.⁹⁵ Amyloid deposits may produce isolated involvement of the ureter. Furthermore, a pelvic mass or inflammatory process associated with actinomycosis may cause external ureteral compression.^{96,97} Retrovesical echinococcal cyst can also impede urine flow.⁹⁸ Retroperitoneal malacoplakia can also be a rare cause of urinary obstruction.⁹⁹ Polyarteritis nodosa associated with hepatitis B has also been reported to result in bilateral hydronephrosis.¹⁰⁰

Hematologic abnormalities induce obstruction of the urinary tract by a variety of mechanisms. In the retroperitoneum, enlarged lymph nodes or a tumor mass may compress the ureter.^{33,101} Alternatively, precipitation of cellular breakdown products such as uric acid (see discussion earlier) and paraproteins, as in multiple myeloma, may cause intrinsic obstruction. In patients with clotting abnormalities, blood clots or hematomas may obstruct the urinary tract, as can sloughing of the papillae in patients with sickle cell disease or analgesic nephropathy (see discussion earlier). Although leukemic infiltrates rarely cause obstruction in adults, in children they cause obstruction in 5% of patients.¹⁰²

Lymphomatous infiltration of the kidney occurs relatively commonly, but obstruction related to ureteral involvement in lymphoma is rarer.¹⁰³

CLINICAL ASPECTS

Urinary tract obstruction may cause symptoms referable to the urinary tract. However, even patients with severe obstruction may be asymptomatic, especially in settings where the obstruction develops gradually, or in patients with spinal cord injury.¹⁰⁴ The clinical presentation often depends on the rate of onset of the obstruction (acute or chronic), the degree of obstruction (partial or complete), whether the obstruction is unilateral or bilateral, and whether the obstruction is intrinsic or extrinsic. Pain in obstructive uropathy is usually associated with obstruction of sudden onset, as from a kidney stone, blood clot, or sloughed papilla, and appears to result from abrupt stretching of the renal capsule or the wall of the collecting system, where C-type sensory fibers are located. The severity of the pain appears to correlate with the rate, rather than the degree, of distention. The pain may present as typical renal colic (sharp pain that may radiate toward the urethral orifice), or, in patients with reflux, the pain may radiate to the flank only during micturition. With UPJ obstruction, flank pain may develop or worsen when the patient ingests large quantities of fluids or receives diuretics.¹⁰⁵ Early satiety and weight loss may be another symptom.¹⁰⁶ Ileus or other gastrointestinal symptoms may be associated with the pain, especially in cases of renal colic, so that it can be difficult to differentiate obstruction from gastrointestinal disease.

Sometimes, patients notice changes in urine output as obstruction sets in. Urinary tract obstruction is one of the few conditions that can result in anuria, usually because of bladder outlet obstruction, or obstruction of a solitary kidney at any level. Obstruction may also occur with no change in urine output. Alternatively, episodes of polyuria may alternate with periods of oliguria. Recurrent urinary tract infections may be the only sign of obstruction, particularly in children. As mentioned earlier, prostatic disease with significant bladder outlet obstruction often presents with difficulty initiating urination, decreased size or force of the urine stream, postvoiding dribbling, and incomplete emptying.¹⁰⁷ Spastic bladder or irritative symptoms such as frequency, urgency, and dysuria may result from urinary tract infection. The appearance of obstructive symptoms synchronous with the menstrual cycle may also be a sign of endometriosis.¹⁰⁸

On physical examination, several signs may suggest urinary obstruction. A palpable abdominal mass, especially in neonates, may represent hydronephrosis, or, in all age groups, a palpable suprapubic mass may represent a distended bladder. On laboratory examination, proteinuria, if present, is generally less than 2 g/day. Microscopic hematuria is a common finding, but gross hematuria may develop occasionally, such as in rare cases with appendiceal granuloma.¹⁰⁹ The urine sediment is often unremarkable. Less common manifestations of urinary tract obstruction include deterioration of renal function without apparent cause, hypertension,¹¹⁰ polycythemia, and abnormal urine acidification and concentration capacity.

DIAGNOSIS

Careful history and physical examination represent the cornerstone of diagnosis, often leading to detection of urinary tract obstruction, and suggesting the reason for it. Although challenging, history and physical examination should focus on the evaluation, so that the minimum amount of time and expense are incurred in determining the cause of the obstruction.

HISTORY AND PHYSICAL EXAMINATION

Important information in the history includes the type and duration of symptoms (voiding difficulties, flank pain, decreased urine output), presence or absence of urinary tract infections and their number and frequency (especially in children), pattern of fluid intake and urine output, as well as any symptoms of chronic renal failure (such as fatigue, sleep disturbance, loss of appetite, pruritis). In addition, relevant medical history should be reviewed in detail, looking for predisposing causes, including stone disease, malignancies, gynecologic diseases, history of recent surgery, AIDS, and drug use.

The physical examination should focus first on vital signs, which may provide evidence of infection (fever, tachycardia), or of volume overload (hypertension). Evaluation of the patient's volume status will guide fluid therapy. The abdominal examination may reveal a flank mass, which may represent hydronephrosis (especially in children), or a suprapubic mass, which may represent a distended bladder. Features of chronic renal failure, such as pallor (anemia), drowsiness (uremia), neuromuscular irritability (metabolic abnormalities), or pericardial friction rub (uremic pericarditis), may also be noted. A thorough pelvic examination in women and a rectal examination for all patients are mandatory. A careful history and a well-directed and complete physical examination often reveal the specific cause of urinary obstruction. Coexistence of obstruction and infection is a urologic emergency and appropriate studies (US, CT, MRI) must be performed immediately, so that the obstruction can be relieved promptly. Intravenous urography (IVU) is nowadays less commonly used in the diagnostic workup of these patients.

BIOMARKERS FOR EVALUATION OF URINARY TRACT OBSTRUCTION

The effects of chronic obstruction on renal function are the result of a complex series of events that profoundly and progressively alter all components of glomerular and tubular functions. The consequence is development of tubulointerstitial injury that is characterized by dynamic changes involving tubular atrophy, inflammatory cell infiltration, and interstitial fibrosis. This leads to irreversible loss of kidney function and obstructive nephropathy, which even may continue to progress after relief of obstruction. Because the complexity of this process involves almost every cell in the kidney and many pathways, there have been many attempts to identify biomarkers, which predict the course of obstruction. These biomarkers are characterized by the current understanding of the pathophysiology of urinary tract obstruction on kidney injury and currently include biochemical

assays of blood and urine as well as many imaging tools as detailed later.

BIOCHEMICAL EVALUATION OF BLOOD AND URINE

The laboratory evaluation includes urinalysis and examination of the sediment on a fresh specimen by an experienced observer. Unexplained renal failure with benign urinary sediment should suggest urinary tract obstruction. Microscopic hematuria without proteinuria may suggest calculus or tumor. Pyuria and bacteriuria may indicate pyelonephritis; bacteriuria alone may suggest stasis. Crystals in a freshly voided specimen should lead to consideration of nephrolithiasis or intrarenal crystal deposition.

Hematologic evaluation includes the hemoglobin/hematocrit and mean corpuscular volume (to identify anemia of chronic renal disease) and white blood cell count (to identify possible hematopoietic system neoplasm or infection). Serum electrolytes (Na^+ , Cl^- , K^+ , and HCO_3^-), blood urea nitrogen concentration, creatinine, Ca^{2+} , phosphorus, Mg^{2+} , uric acid, and albumin levels should be measured. These will help identify disorders of distal nephron function (impaired acid excretion or osmoregulation) and uremia. Urinary chemistries may also suggest distal tubular dysfunction (high urine pH, isosthenuric urine) and inability to reabsorb sodium normally (urinary $\text{Na}^+ > 20$ mEq/L, fractional excretion of Na^+ [FE_{Na}] $> 1\%$, and osmolality < 350 mOsm/L). Alternatively, in acute obstruction, urinary chemistry values may be consistent with prerenal azotemia (urinary $\text{Na}^+ < 20$ mEq/L, $\text{FE}_{\text{Na}} < 1\%$, and osmolality > 500 mOsm/L).⁸

Novel biomarkers relevant for the functional as well as cellular and molecular changes are being developed as an index of renal injury and to predict renal reserve or recovery after reconstruction. Simple tests examining the value of risk factor proteins have suggested that elevated levels of neutrophil gelatinase-associated lipocalin and $\beta 2$ -microglobulin are present in the urine from obstructed kidneys.¹¹¹ Attempts to predict the clinical outcome of congenital unilateral UPJ obstruction in newborn by urine proteome analysis reveal an example of this powerful new technology. Polypeptides in the urine were identified and enabled diagnosis of the severity of obstruction, and using this technique the clinical evolution was predicted with 94% precision in neonates,¹¹² whereas the precision was only 20% in older children with UPJ obstruction.¹¹³ Thus far, application of large-scale urinary proteomic analysis holds promise for better classification of individuals with hydronephrosis for early selection of surgical candidates,¹¹⁴ but long-term follow-up studies are warranted to determine the true clinical value of this diagnostic approach.¹¹⁵ Posterior urethral valves are diagnosed by antenatal US and by analyzing the fetal urinary peptidome; a classifier based on 12 fetal urine peptides predicted postnatal renal function with high sensitivity and specificity.¹¹⁶

In an attempt toward improved understanding of obstructive nephropathy and improved translatability of the results to clinical practice, a systems biology approach combining omics data of both human and mouse obstructive nephropathy was developed and demonstrated that novel markers can be identified only through a translational approach. Using the urinary miRNome of infants with UPJ obstruction and the kidney tissue miRNome and transcriptome of the

corresponding neonatal partial unilateral ureteral obstruction (PUUO) mouse model revealed that let-7a and miR-29b are potentially involved in the development of fibrosis in UPJ obstruction via the control of DTX4 in both humans and mice.¹¹⁷

EVALUATION BY MEDICAL IMAGING

The history, physical examination, and initial laboratory studies should guide the medical imaging evaluation. Pain, degree of renal dysfunction, and the presence of infection dictate the speed and nature of the evaluation. Numerous imaging techniques are available; each has its own advantages and disadvantages, including the ability to identify the site and cause of the obstruction and to separate functional obstruction from mere dilation of the urinary tract. Patient-specific factors, such as the risk of radiocontrast in the setting of renal insufficiency, or the risk of exposure to radiation in pregnant women, must also be weighed.²⁰

ULTRASONOGRAPHY

Ultrasonography (US) is the preferred screening modality when obstruction is suspected^{118,119} because it is highly sensitive for hydronephrosis,^{118,119} is safe and can be repeated frequently, is readily available at low cost, and avoids ionizing radiation, thus making it ideal for pregnant patients,²⁰ infants, and children.^{118,120} Moreover, because US requires no radiographic contrast, it is well suited to rule out obstruction as a cause of renal insufficiency in patients in whom contrast is contraindicated, including those with an elevated or rising serum creatinine level,^{118,119} those allergic to contrast material, and in pediatric patients.¹²¹ In addition to detecting hydronephrosis, US can reveal dilatation of the renal pelvis and calices. It may also determine the size and shape of the kidney, and may demonstrate thinned cortex in case of severe long-standing hydronephrosis (Fig. 37.1). Finally, US may detect perinephric abscesses, which may complicate some forms of obstructive nephropathy. Importantly, in a recent multicenter comparative effectiveness trial in patients with suspected nephrolithiasis subjected to either point of care ultrasonography (emergency US), US performed by a radiologist, or abdominal CT, it was concluded that using ultrasonography as the initial test resulted in no need for CT in most patients, lower cumulative radiation exposure, and no significant differences in the risk of subsequent serious adverse events, pain scores, return emergency department visits, or hospitalizations.¹²²

Ultrasonography is both highly sensitive and highly specific in detecting hydronephrosis, with the rates approaching 90%.^{118,119,123,124} Importantly, US works equally well in patients with azotemia, in whom radiocontrast studies are contraindicated.¹²⁴ Hydronephrosis is detected as a dilated collecting system—an anechoic central area surrounded by echogenic parenchyma.

However, in some cases of acute urinary obstruction, US may fail to detect pathology. During the first 48 hours of obstruction,^{118,119,123,124} or when hydronephrosis is absent despite obstruction evaluated by CT, the US may reveal no abnormality.¹²⁵ False-negative results also occur in cases of dehydration, staghorn calculi, nephrocalcinosis,¹²⁴ retroperitoneal fibrosis,¹²⁶ misinterpretation of caliectasis as cortical cysts,¹²⁷ and in cases of tumor encasement of the

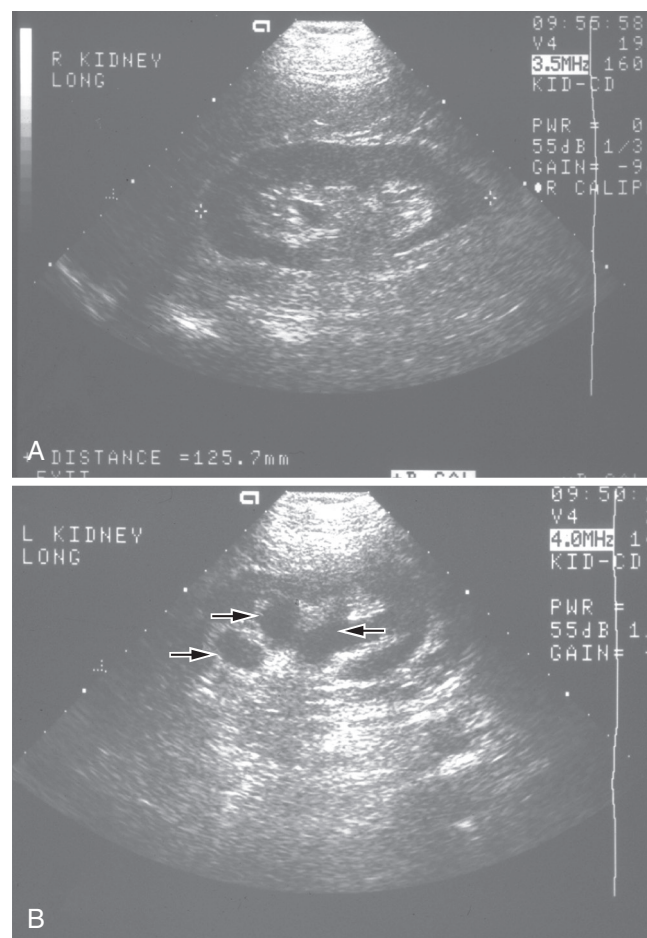


Fig. 37.1 Renal ultrasound. (A) Normal kidney. (B) Hydronephrotic kidney: dilated calices and pelvis (arrows).

collecting system.¹²⁸ A dilated collecting system without obstruction may be observed in up to 50% of patients with urinary diversion through ileal conduits.¹²⁹ To enhance the sensitivity and specificity of US, some investigators have developed special obstructive scoring systems, which grade increased echogenicity, parenchymal rims greater than 5 mm, contralateral hypertrophy, resistive index (RI) ratio of 1.10 or higher, and other features to differentiate between obstructing and nonobstructing hydronephrosis.¹²⁰ False-positive studies may result from a large extrarenal pelvis, parapelvic cysts,¹³⁰ vesicoureteral reflux, or high urine flow rate.¹²⁴ In addition, US may only suggest, but not reveal, the presence, or cause, of the obstruction. Renal US elastography provides measurement of kidney elasticity by the shear wave technique.¹³¹ This new imaging technique provides information about renal stiffness related to fibrosis (i.e., chronic obstruction). However, elastography is also sensitive to mechanical and functional parameters such as hydronephrosis and external pressure.¹³¹

Importantly, although US is a useful screening test, it does not define renal function and cannot completely rule out obstruction, especially when prior clinical suspicion is high. Every experienced nephrologist has seen cases of obstruction with negative US studies. Therefore the diagnosis of obstruction must still be considered in patients with worsening renal function, chronic azotemia, or acute changes in renal function

or urine output, even in the absence of hydronephrosis on the US.¹³²

ANTENATAL ULTRASONOGRAPHY

Prenatal diagnosis of renal pathology was first described in the 1970s.¹³³ After that, routine maternal ultrasonography devices of ever-increasing resolution resulted in a fourfold increase in antenatal detection of congenital urinary tract obstruction.¹³⁴ Prenatal hydronephrosis is diagnosed with an incidence of between 1 in 100 and 1 in 500 in maternal–fetal US studies.^{10,12,102} Either obstructive or nonobstructive processes can cause dilation of the urinary tract. Overall, the etiology of urinary tract obstruction includes UPJ obstruction (44%), or the ureterovesical junction obstruction (21%), as well as multicystic dysplastic kidney, ureterocele or ureteral ectopia, duplex kidney (12%), posterior urethral valves (9%), urethral atresia, sacrococcygeal teratoma, and hydrometrocolpos (fluid distention of the uterus).^{103,135–137} Nonobstructive causes include vesicoureteral reflux (14%), physiologic dilation, prune-belly syndrome, renal cystic disease, and megacalycoses (massive dilation of the renal calyces).^{103,135–137} Increased renal echogenicity and oligohydramnios (inadequate quantities of amniotic fluid) in the setting of bladder distention are highly predictive (87%) of an obstructive etiology. This finding is important in the prenatal counseling and treatment of boys with bilateral hydronephrosis and marked bladder dilation.¹³⁸

Determining which cases require intervention and which can be treated conservatively remains a major issue in prenatal US diagnosis of urinary tract obstruction. Persistent postnatal renal abnormalities appear likely when the anteroposterior diameter of the fetal renal pelvis measures more than 6 mm at less than 20 weeks, more than 8 mm at 20 to 30 weeks, and more than 10 mm at more than 30 weeks of gestation. The long-term morbidity of mild hydronephrosis (pelviectasis without caliceal dilation) is low.^{10,12} Moderate hydronephrosis (dilated pelvis and calices without parenchymal thinning) may be associated with gradual improvement in severity of dilation, without loss of anticipated relative renal function. Cases of severe hydronephrosis (pelviccalyceal dilation with parenchymal thinning) may require surgical intervention for declining renal function, infection, or symptoms. Overall, because approximately only 5% to 25% of patients with antenatal hydronephrosis will ultimately require surgical intervention,^{102,139} careful long-term follow-up of these patients is required throughout childhood and into adulthood. Almost all patients with antenatal hydronephrosis will have postnatal ultrasonography performed in the first days of life, keeping in mind that most cases of the mild hydronephrosis will resolve without intervention.¹⁴⁰ Functional imaging is required to define residual renal function of patients with hydronephrosis and to monitor its course over postnatal life. However, in the absence of bilateral hydronephrosis, a solitary kidney, or suspected posterior urethral valve, functional imaging can be deferred until the first 4 to 6 weeks of life.¹⁰² Otherwise, nuclear medicine examination with radioisotope renography should be performed.

In the United States most infants with prenatally detected hydronephrosis that is confirmed with postnatal studies are placed on antibiotic prophylaxis pending the outcome of further evaluation.¹⁰² This is not the routine treatment in Europe. However, an infection in the setting of ureteral

obstruction can cause significant morbidity, resulting in an infant with sepsis, and renal damage is a potential comorbidity. Oral amoxicillin (10 mg/kg/day) is the most commonly used prophylactic antibiotic.¹⁰²

DUPLEX DOPPLER ULTRASONOGRAPHY

As detailed earlier, US is very sensitive for the detection of collecting system dilatation (hydronephrosis); however, obstruction is not synonymous with dilatation, as either obstructive or nonobstructive dilatation may be present. To differentiate these conditions, color duplex Doppler with measurement of the RI in the intrarenal arteries may be helpful, as obstruction (except in the acute and subacute stages) leads to intrarenal vasoconstriction with a consecutive increase of the RI above the upper limit of 0.7, whereas nonobstructive dilatation does not.^{103,135} Diuretic challenge to the kidney may further enhance these differences in RI between obstruction and dilatation.¹³⁶ Of clinical relevance in one study, renal colic comprised 30% to 35% of all urological emergencies, and color Doppler predicted the onset of acute dilatation with higher sensitivity, specificity, accuracy, and diagnostic efficiency than ultrasonography in patients with renal colic.¹⁴¹

INTRAVENOUS UROGRAPHY

IVU (also known as intravenous pyelography) was for many years state-of-the-art when a patient suspected with a history of urinary tract obstruction was referred for imaging (Fig. 37.2). IVU requires administration of radiological contrast and is time-consuming to perform. It has therefore now been largely replaced by US, CT, and MRI.



Fig. 37.2 Intravenous pyelography. Normal right kidney and dilated collecting system on the left. The obstruction was relieved with a stent.

COMPUTED TOMOGRAPHY

CT was initially used mainly in cases with a high index of clinical suspicion, in which US or IVU had failed to identify obstruction.¹⁴² With the advance to higher resolution of multidetector row CT scanners, the CT scan has supplanted IVU for evaluation of the upper urinary tract.^{142,143} CT has a particular advantage because it can visualize a dilated collecting system without the requirement for contrast enhancement. It can also be performed much more quickly than IVU, especially when renal impairment or obstruction would delay contrast excretion by the affected kidney in an IVU (Fig. 37.3). Noncontrast-enhanced CT identifies ureteral stones more effectively than IVU and detects the presence or absence of ureteral obstruction as effectively as IVU.^{144,145} Because of its exquisite sensitivity to density, CT can identify even radiolucent stones, because even uric acid stone density is at least 100 Hounsfield units (HU), which is higher than soft tissue density on CT (usually 10–70 HU). CT is especially effective in identifying extrinsic causes of obstruction (e.g., retroperitoneal fibrosis, lymphadenopathy, hematoma). The use of CT for the diagnosis of stones has increased by a factor of 10 over the past 15 years in the United States,¹⁴⁶ probably because of its greater sensitivity and

because it can be performed as needed in most emergency departments in the United States.¹⁴⁷ This was highlighted in a recent multicenter comparative effectiveness trial in patients with suspected nephrolithiasis subjected to either point of care ultrasonography (emergency US), US performed by a radiologist, or abdominal CT, which demonstrated that CT has a higher sensitivity than US, whereas US has higher specificity than CT.¹²² In this regard it must be emphasized that although the cancer risk from radiation exposure to CT scans is very low, CT scans might produce a small additional cancer risk,¹⁴⁸ especially in those subjected to CT scans during childhood.¹⁴⁹

Helical CT has also proven to be an accurate and noninvasive method of demonstrating crossing vessels in UPJ obstruction.¹⁵⁰ CT can detect extraurinary pathology and can establish nonurogenital causes of pain. All of these advantages establish noncontrast-enhanced helical CT as the diagnostic study of choice for the evaluation of the patient with acute flank pain.¹²⁵ CT is very useful in delineating the pelvic organs, such as the bladder and prostate, and may demonstrate abnormalities such as an obstructed and distended bladder (Fig. 37.4) secondary to an enlarged prostate. US may be the first method of diagnosis in this setting (Fig. 37.5), but CT resolution and depiction of details are usually

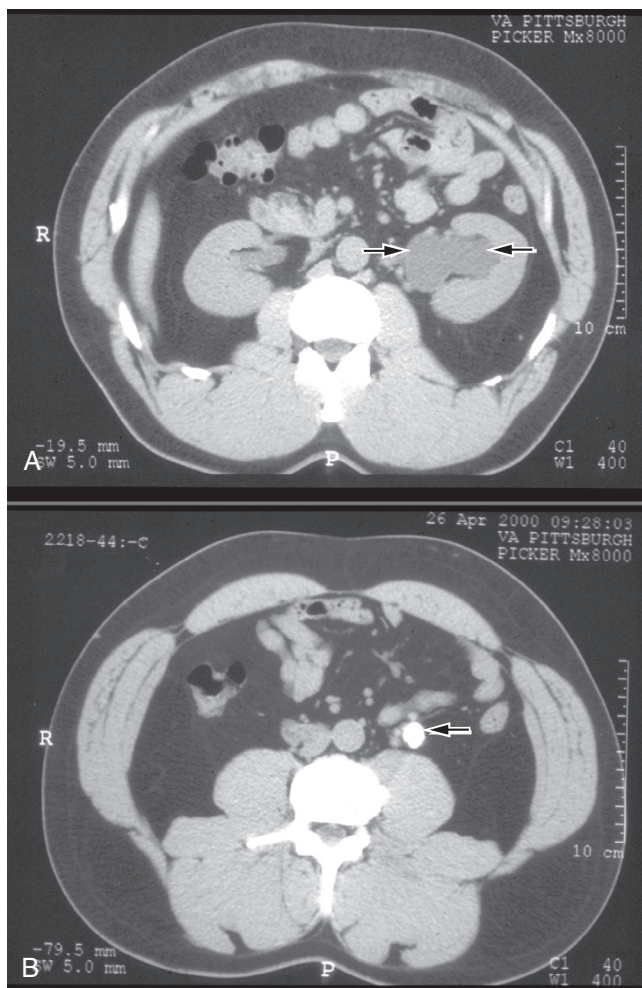


Fig. 37.3 Computed tomography, noncontrast study. (A) Left hydronephrosis: dilated renal pelvis (arrows), with normal kidney on the right. (B) Reason for obstruction: left midureteral stone (arrow).

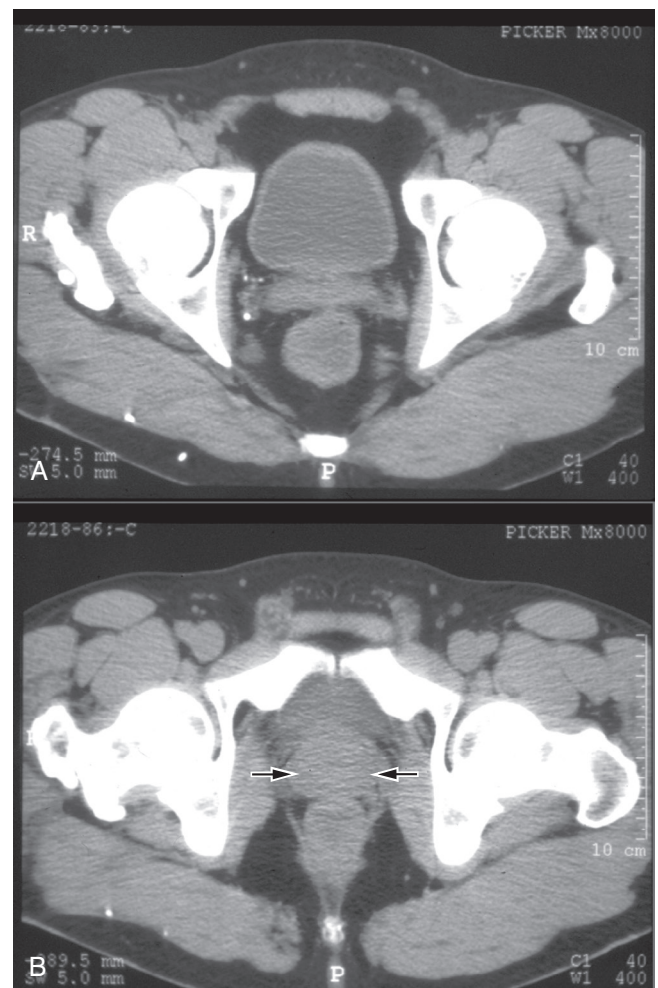


Fig. 37.4 Computed tomography of the pelvis. (A) Large postvoiding residual urine in the bladder. (B) Enlarged prostate (arrows), leading to urinary retention.

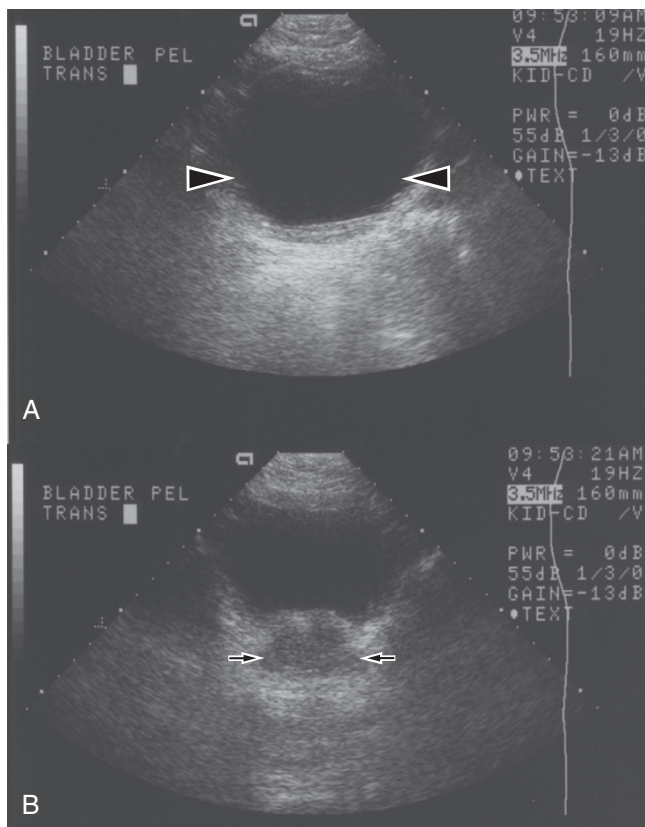


Fig. 37.5 Pelvic ultrasound. (A) Distended bladder (arrowheads). (B) Enlarged prostate (arrows), causing infravesical urinary obstruction.

superior to those of US.¹⁴² An exception to using a noncontrast CT is nephrolithiasis secondary to human immunodeficiency virus (HIV) protease inhibitors, primarily indinavir. These stones are not radiopaque, and signs of obstruction may be minimal or absent; thus the diagnosis may be missed with US and noncontrasted CT scan. Contrast-enhanced CT scanning may be required to establish the diagnosis in this circumstance.¹⁵¹

ISOTOPIC RENOGRAPHY

Isotopic renography, or renal scintigraphy, is helpful in diagnosing upper urinary tract obstruction and providing information on the differential renal function (DRF) of both kidneys, while avoiding the risk of radiocontrast agents.^{152,153} Radioisotope is injected intravenously, and its dynamic uptake and excretion by the kidneys are followed by using imaging with a gamma camera. Although this method gives a functional assessment of the obstructed kidney, anatomic definition is suboptimal compared with CT. Isotopic renography is typically used to estimate the fractional contribution of each kidney to overall renal function. The noninvasive character of this examination with its high reproducibility makes it excellent for monitoring patients, and it helps the urologist decide whether to perform surgical intervention or watchful waiting.²³ In addition, the test can be repeated after the relief of obstruction to gauge the extent to which relief of the obstruction has restored renal function.

Diuretic renography was introduced into clinical practice in 1978,¹⁵⁴ and may be used to distinguish between hydronephrosis or pelvic dilation with obstruction and dilation

without obstruction. The method was developed, applied, and validated in adults.¹⁵⁴ In particular, this is important when applying diuretic renography in children and infants where it is not always easy to distinguish between dilation and obstruction. Following administration of radioisotope, when the isotope appears in the renal pelvis, a loop diuretic such as furosemide is given intravenously. If stasis is causing the dilation, the induced diuresis may result in prompt washout of the tracer from the renal pelvis. By contrast, when dilation is caused by obstruction, the washout does not occur.¹⁵⁵ Data should be interpreted visually and by quantitative measurement including the half-life ($T_{1/2}$) for the excretion of the tracer from the collecting system.¹⁵⁶ It is generally accepted that the clearance of the isotope from the collecting system with $T_{1/2}$ less than 15 minutes is normal, and a $T_{1/2}$ of more than 20 minutes may indicate obstruction in adults. Renal excretion of the tracer with a $T_{1/2}$ between 15 and 20 minutes is considered equivocal. An absent or blunted diuretic response resulting from decreased renal function or grossly dilated pelvis makes interpretation of the test difficult and limits its usefulness and may require support tools to increase the diagnostic performance.¹⁵⁷ Moreover, in children diuretic renography is a very important method for guiding the management of asymptomatic congenital hydronephrosis. The classical variables of the diuretic renogram may not allow an estimate of the best drainage. Poor pelvic emptying may be apparent because the bladder is full and because the effect of gravity on drainage is incomplete. Estimating the drainage as residual activity rather than any parameter on the slope might be more adequate, especially if the time of furosemide administration is changed. Renal function and pelvic volume can influence the quality of drainage. Misinterpretation can easily take place if the renal pelvis is very large and does not allow proper drainage within the study time and drainage may be better estimated using new tools.¹⁵⁸ From this examination the DRF can also be obtained, which is a robust measure, provided there is adequate background subtraction. Pitfalls are related to the drawing of regions of interest, particularly in infants, to estimating the interval during which DRF is calculated, and to an adequate signal-to-noise ratio. There is no definition of a “significant” reduction in DRF.

POSITRON EMISSION TOMOGRAPHY

Experimentally, positron emission tomography permits quantitative imaging of the kidney at a spatial resolution appropriate for the organ. $H_2^{15}O$, $^{82}RbCl$, and $[^{64}Cu]$ ETS (ethylglyoxal bis[thiosemicarbazone]) are the most important radiopharmaceuticals for measuring renal blood flow.¹⁵⁹ Membrane organic cation and anion transporters are important for the function of the tubular epithelium and novel radiopharmaceuticals, such as copper-64-labeled mono oxo-tetraazamacrocyclic ligands and carbon-11-labeled metformin, have been used for molecular renal imaging^{159,160} and potentially may play a role in detection of kidney injury in response to urinary tract obstruction.¹⁶¹

MAGNETIC RESONANCE IMAGING

New MRI systems and specific MR contrast agents provide significant developments in the evaluation of renal function (glomerular filtration rate [GFR] measurement), assessment of potential prognostic factors (hypoxia, inflammation, cell



Fig. 37.6 Magnetic resonance (MR) imaging urography of left-sided hydronephrosis with parenchymal thinning. MR urographic image shows large dilatation of the left pelvicalyceal system and narrowing of the left ureteropelvic junction segment.

viability, degree of tubular function, and interstitial fibrosis), and for monitoring new therapies.^{143,162} New developments that have provided higher signal-to-noise ratio and higher spatial and/or temporal resolutions have the potential to direct new opportunities for obtaining morphologic and functional information on tissue characteristics that are relevant for various renal diseases including urinary tract obstruction with respect to diagnosis, prognosis, and treatment follow-up.^{143,162} MRI can be used to explore the urinary tract when obstruction is suspected. MRI provides improved spatial resolution, and it is superior to IVU in detecting obstruction in the presence of severe renal failure.¹⁴³ MRI has very limited application for the evaluation of stone disease because it cannot directly detect calcifications or stone material.¹⁶³

Depending on local conditions, MRI may be more expensive than other modalities. In children, MR urography (Fig. 37.6) may replace conventional urological methods, and a recent study suggests that functional MR plays an important potential role in identifying those who will benefit most from pyeloplasty and those who are probably best observed.¹⁶⁴ Promising experimental studies have recently demonstrated that MRI may provide valuable information regarding renal function, including energy consumption from so-called BOLD (blood oxygenation level dependent) imaging; this kind of data may be helpful in the future in predicting the level of return of renal function following obstruction.^{165,166} Of interest, functional MRI using BOLD demonstrated the ability to identify pathophysiological changes in patients with acute obstruction due to calculi in the ureter.¹⁶⁷ Using new BOLD-MRI analysis techniques, it has recently been shown that persons suffering from chronic kidney disease (CKD) have lower cortical oxygenation than normotensive controls, thus confirming the chronic hypoxia hypothesis. The acute alterations in BOLD after the administration of

furosemide are smaller in CKD and represent an estimate of the oxygen-dependent tubular transport of sodium. BOLD-MRI alone or in combination with other functional MRI methods can thus be used to monitor the renal effects of drugs and may potentially be used to identify parenchymal changes in response to urinary tract obstruction. The near future will tell whether or not BOLD-MRI represents a new tool to predict renal function decline and adverse renal outcomes.¹⁶⁸ Recently, concerns related to the use of MRI in renal patients were highlighted by the increased risk of developing nephrogenic systemic fibrosis (NSF) induced by the toxicity of gadolinium in patients with severely impaired renal function, whereas patients with normal or moderate renal function impairment do not develop NSF. It is recommended that gadolinium contrast media should be avoided in patients with stage 4 or 5 CKD because of the risk of NSF.¹⁶⁹

WHITAKER TEST

The Whitaker test traditionally defines the functional effect of upper urinary tract dilatation by measuring the hydrostatic pressures in the renal pelvis and bladder during infusion of a saline and contrast mixture into the renal pelvis, via a catheter.¹⁷⁰ With a bladder catheter in place, the patient is placed in the prone position on the fluoroscopic table and a cannula is inserted percutaneously into the renal pelvis and connected to a pressure transducer. A mixture of saline and contrast material is infused through the renal cannula at a rate of 10 mL/min, and pressures are monitored. The urinary tract is considered nonobstructed if renal pelvic pressure is less than 15 cm H₂O, equivocal at a pressure between 15 and 22 cm H₂O, and obstructed if pressure exceeds 22 cm H₂O.¹⁷⁰ With the advent of noninvasive imaging techniques, this test should be reserved for assessing potential upper urinary tract obstruction only in the following circumstances: equivocal results from less invasive tests, suspected obstruction with poor kidney function, loin pain with a negative diuresis renogram, suspected intermittent obstruction, and gross dilatation with a positive diuresis renogram.¹⁷¹ The test is only used rarely and interpretation relies on solid experience with the method.

RETROGRADE AND ANTEGRADE PYELOGRAPHY

When other tests do not provide adequate anatomic detail, or when obstruction must be relieved (e.g., obstruction of a solitary kidney, bilateral obstruction, or symptomatic infection in the obstructed system), more invasive investigation, with a combination of treatments, may be necessary. When retrograde pyelography is performed, this takes place during cystoscopy, by cannulating the ureteral orifice and injecting contrast.^{61,172,173} In some cases of complete obstruction, contrast may not reach the kidney, but the procedure will define the lower level of the obstruction. Retrograde pyelography can be combined with placement of a ureteral stent to relieve an obstruction, or with possible stone extraction. Because the catheter passes through the bladder to reach the upper urinary tract, the risk of introducing infection proximal to the obstruction must be kept in mind, and the obstruction should be relieved immediately after retrograde pyelography. Antegrade pyelography is performed by percutaneous cannulation of the renal pelvis, and injection of the contrast material into the kidney and ureter.^{172,173} This procedure should establish the proximal level of obstruction and may

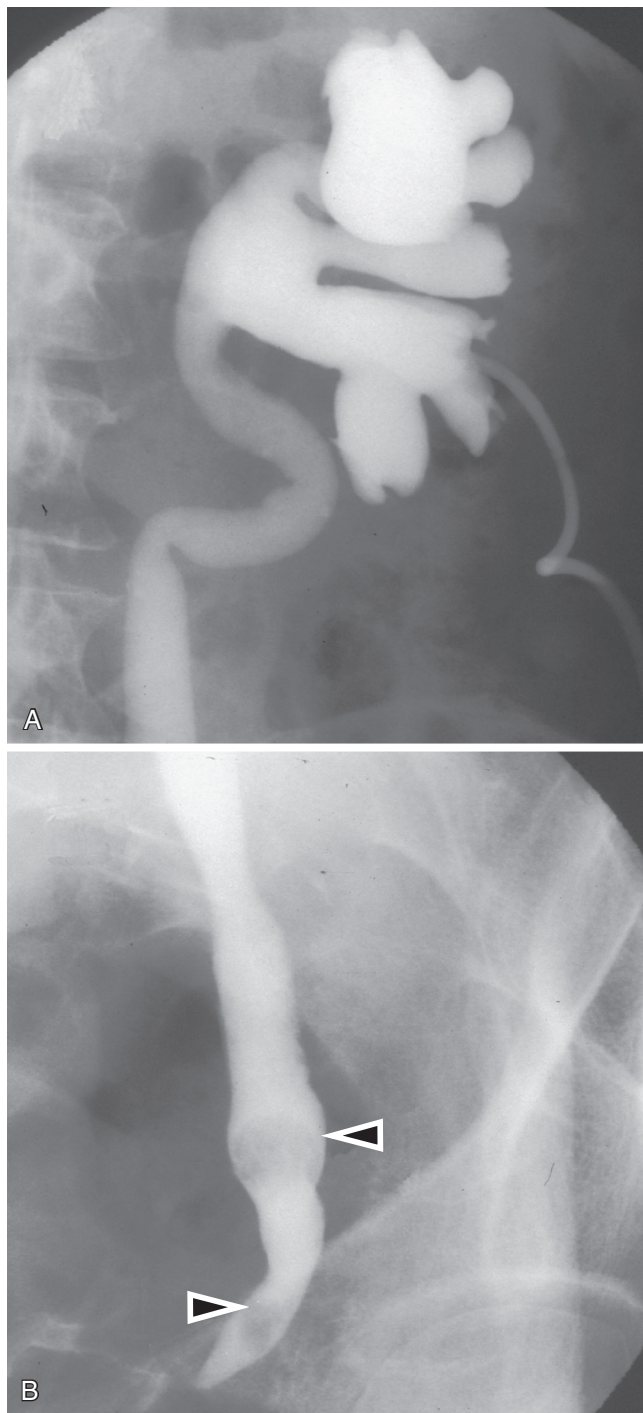


Fig. 37.7 Antegrade pyelography. (A) Dilated renal pelvis and calices on left. (B) Stones (arrowheads) as filling defects in the distal ureter (not seen on plain film). Intravenous pyelography was unsuccessful owing to the obstructed and malfunctioning kidney.

also serve as a first step in relieving obstruction by means of percutaneous nephrostomy (Fig. 37.7).

PATHOPHYSIOLOGY OF OBSTRUCTIVE NEPHROPATHY

Despite the fact that acquired obstructive nephropathy in humans usually results from partial urinary tract obstruction

and is generally prolonged in its time course, most mechanistic studies of renal dysfunction in acquired obstruction use models of acute complete obstruction, usually for 24 hours. In these animal models, the extent of obstruction is clear and reproducible, and, if the kidneys are studied soon after the obstruction is performed or released, the results are not confounded by changes in renal structure brought on by inflammation or fibrosis. Complete obstruction of short duration strikingly alters renal blood flow, glomerular filtration, and tubular function, while producing minimal anatomic changes in blood vessels, glomeruli, and tubules.²

EFFECTS OF OBSTRUCTION ON RENAL BLOOD FLOW AND GLOMERULAR FILTRATION

Obstruction profoundly alters all components of glomerular function. The extent of the disturbance in GFR depends on the severity and duration of the obstruction, whether it is unilateral or bilateral, and the extent to which the obstruction has been relieved or persists.² To describe the effects of obstruction on glomerular filtration, we must review aspects of normal GFR. Whole-kidney GFR depends on the filtration rate of all functioning glomeruli and the proportion of glomeruli actually filtering. As detailed in Chapter 3, single-nephron GFR (SNGFR) is determined by the blood flow in the glomerulus, the net ultrafiltration pressure across the glomerular capillary, and the ultrafiltration coefficient (K_f). Glomerular blood flow and the hydraulic pressure in the glomerular capillary (P_{GC}) are determined by the resistances of the afferent (R_A) and efferent (R_E) arterioles. Net ultrafiltration pressure is determined by P_{GC} , the hydraulic pressure of the Bowman space (which equals the proximal tubule hydraulic pressure, P_T), and the differences in oncotic pressure between the glomerular capillary and Bowman space. K_f is determined by the permeability properties of the filtering surface and the surface area available for filtration. Obstruction can alter one or all of these determinants of GFR.

THE EARLY, HYPEREMIC PHASE

In the immediate 2 to 3 hours following the onset of UUO, blockade of antegrade urine flow markedly increased P_T . This increase in pressure in Bowman space would be expected to halt GFR immediately.^{174–176} However, during this early phase of obstruction, the afferent arterioles dilate, decreasing R_A , increasing P_{GC} , and counteracting the increase in P_T .^{174,175} Because this vasodilator or “hyperemic response” occurs in denervated kidneys in situ and in isolated perfused kidneys,^{177,178} it must result from intrarenal mechanisms. In fact, glomeruli of individual nephrons exhibit the same response in in vivo micropuncture experiments when antegrade urine flow is blocked by placement of a wax block in the tubule of the nephron.¹⁷⁹

Many mechanisms may mediate this afferent vasodilation, including increases in vasodilator hormones such as prostaglandins, regulation by the macula densa, and a direct myogenic reflex. This hyperemic response is not attenuated by renal nerve stimulation or infusion of catecholamines,¹⁸⁰ and it may be linked to changes in interstitial pressure.¹⁷⁸

In the tubuloglomerular feedback response, reduced tubular flow past the macula densa induces reductions in R_A and increases in P_{GC} , so that SNGFR rises. Similarly, because obstruction reduces urine flow past the macula densa, it

would be predicted to induce afferent vasodilation.¹⁷⁹ However, elegant micropuncture studies separated the stoppage in flow from increases in P_T by placing an additional puncture in the tubule that was proximal to the blockage of flow to the macula densa. In this setting, flow past the macula densa was halted, but P_T remained normal, because accumulating tubular fluid was permitted to leak out.¹⁷⁵ In such nephrons the increase in P_{GC} observed in obstructed tubules did not occur, indicating that the obstruction itself and not the macula densa stimulates afferent vasodilation.¹⁷⁵

Renal prostaglandins and renal nerves play important roles in the hyperemic response. Indomethacin blocks the hyperemic response, indicating that vasodilator prostaglandins are critical to afferent vasodilation.^{176,181} A renorenal reflex mechanism in the hemodynamic response to obstruction can be discerned from studies in bilateral obstruction, where the afferent vasodilation response is absent or markedly attenuated.^{2,178} Obstruction of the left kidney augments afferent renal nerve activity from the left kidney and efferent nerve activity to the right kidney. Increased efferent nerve activity to the right kidney was accompanied by reduced blood flow to that kidney. This vasoconstrictor response was ablated by denervation of either the left or right kidney before induction of left ureteral obstruction, suggesting that increased afferent renal nerve traffic triggers vasoconstrictive renorenal reflex activity that counteracts the early intrinsic renal vasodilator effects of obstruction in bilateral ureteral obstruction.¹⁷⁸

THE LATE, VASOCONSTRICTIVE PHASE

Because obstruction results in cessation of glomerular filtration, efforts to study the regulation of SNGFR later in obstruction have measured determinants of GFR immediately after release of obstruction.^{2,182} Using this approach, investigators have shown that renal blood flow declines progressively after 3 hours of unilateral obstruction, and through 12 to 24 hours of obstruction.^{183,184} Interestingly, although tubular pressures rise initially after obstruction, they then decline, so that by 24 hours renal plasma flow, GFR, and intratubular pressures have all dropped below normal values.^{174,176,184,185} At 24 hours into the obstruction, examination of regional blood flow in the kidney by injections of silicone rubber reveals large areas of the cortical vascular bed that are either underperfused or not perfused at all.^{2,176,184} Depending on the species, the different vascular beds in the outer and juxtamedullary cortex receive differing proportions of the renal blood flow under basal conditions and following obstruction. However, it is clear that at 24 hours of obstruction, reduced whole-kidney GFR is due, in large part, to nonperfusion of many glomeruli.

Beyond 24 hours of obstruction, the SNGFR of glomeruli that remain perfused is decreased markedly, both because of reduced blood flow to the afferent arteriole and because of afferent vasoconstriction, which, in turn, reduces P_{GC} .^{185,186} Because P_{GC} responds in the same manner when the individual nephron is blocked with oil for 24 hours before micropuncture measurements are performed, it is clear that afferent arteriolar vasoconstriction plays an important role in attenuating SNGFR during the established phase of obstruction.¹⁸⁷ These results indicate that, like the early hyperemic response, intrarenal mechanisms play the major role in the late vasoconstrictive response to unilateral obstruction. In bilateral obstruction,

Table 37.1 Glomerular Hemodynamics in Ureteral Obstruction^a

Stage of Obstruction	P_T	R_A	P_{GC}	SNGFR
1–2 hours unilateral	↑↑	↓	↑	=
24 hours unilateral	=	↑↑	↓	↓↓
24 hours bilateral	↑↑	=	=	↓↓
After release: 24 hours unilateral	↓	↑↑	↓↓	↓↓
After release: 24 hours bilateral	=	↑↑	↓	↓↓

^aSee text for discussion and references.

P_{GC} , Hydraulic pressure of Bowman space; P_T , proximal tubule hydraulic pressure; R_A , afferent arteriole resistance; SNGFR, single-nephron glomerular filtration rate; =, unchanged; ↑, increased; ↑↑, markedly increased; ↓, reduced; ↓↓, markedly reduced.

renal blood flow is reduced to levels 30% to 60% below normal (Table 37.1).^{185,186,188} In both unilateral and bilateral obstruction, SNGFR falls to a similar degree. However, the mechanisms involved are different in the two conditions. In unilateral obstruction, reduced P_{GC} lowers the driving pressure for filtration when set against a nearly normal P_T . By contrast, in bilateral obstruction, P_{GC} remains normal and GFR is halted by a highly elevated P_T .¹⁸⁵ These results suggest that systemic factors, such as accumulation of extracellular fluid volume and urea, increase in natriuretic substances and alterations in renal nerve activity modulate the vasoconstrictive effect of obstruction on the affected kidney.¹⁸⁸

REGULATION OF THE GLOMERULAR FILTRATION RATE IN RESPONSE TO OBSTRUCTION

The extent to which renal blood flow and GFR are reduced after release of obstruction varies with the species studied and the duration of obstruction.² Following release of a 24-hour complete unilateral obstruction, the GFR remains below 50% of normal in dogs and 25% of normal in rats; renal blood flow remains markedly reduced in both species.² After release of bilateral ureteral obstruction, renal blood flow reaches levels higher than those observed following unilateral obstruction, likely due to systemic natriuretic influences such as volume accumulation, reduced sympathetic tone, or increased circulating atrial natriuretic peptide (ANP), but the GFR remains markedly attenuated. Despite the fact that renal blood flow is increased, GFR remains low in part because of nonperfusion or underperfusion of many glomeruli as shown in silicone rubber injections.^{176,184} Where glomeruli remain perfused, intense afferent vasoconstriction reduces P_{GC} , so that even though P_T also falls with release of the obstruction, the driving force for glomerular filtration remains low.^{185,186} In addition, a sharp reduction in K_f augments the fall in GFR at this point following release of unilateral and bilateral obstruction.^{185,186}

Several mechanisms contribute to afferent vasoconstriction and a reduced K_f . First, release of obstruction strikingly augments the flow of tubular fluid past the macula densa. Although the absolute rate of flow is still far below normal, the macula densa likely senses the dramatic change in the rate of flow, and this may lead to intense vasoconstriction.² In favor of this view, the sensitivity of the tubuloglomerular

feedback mechanism is enhanced in unilateral, as compared with bilateral, obstruction, suggesting that the ability of the mechanism to regulate afferent arteriolar tone is modulated by the extrarenal hormonal milieu.¹⁸⁹

There is substantial evidence that increased intrarenal secretion of angiotensin II (ANG II) participates actively in afferent vasoconstriction and reduced K_f following release of ureteral obstruction. Ureteral obstruction rapidly increases renal vein renin levels at a time when renal blood flow is normal or elevated, but at later time points renal vein renin levels return to normal.^{190–192} In addition, infusion of captopril attenuated the declines in renal blood flow and GFR observed in both unilateral and bilateral obstruction.^{190,192} Because inhibition of angiotensin-converting enzyme can also increase kinin activity, infusions of either carboxypeptidase B, which destroys kinins, or aprotinin, which blocks kinin generation, were used to eliminate the kinin effect. Captopril remained equally effective in the presence of either agent, indicating that captopril reduced R_A primarily by blocking generation of ANG II.² The significance of the renin–angiotensin system as an important contributor to the vasoconstriction has recently been highlighted in studies where AT1 receptor antagonist treatment attenuated the reduction in GFR in the postobstructive period in both adult rats¹⁹³ and rats with neonatally induced unilateral partial obstruction in response to long-term AT1 receptor antagonist treatment.¹⁹⁴

Thromboxane A_2 (TXA₂) plays a role in the obstruction-induced vasoconstriction.^{190,195} Chronically hydronephrotic kidneys exhibit increased TXA₂ accumulation, as measured by accumulation of its more stable metabolite TXB₂.¹⁹⁵ Furthermore, whole-kidney GFR and renal blood flow were increased in response to thromboxane synthase inhibitor treatment,^{190,196} likely by reducing afferent arteriolar resistance and thereby increasing K_f .¹⁹⁷ From these results, TXA₂ appears to be generated in the kidney following release of obstruction and mediates afferent vasoconstriction and reductions in K_f .

Although the source of TXA₂ generation remains unclear in some cases,¹⁹⁸ but not all,¹⁹⁹ glomeruli isolated from obstructed kidneys have shown increased ability to synthesize TXA₂ and other studies have suggested inflammatory cells as the source of TXA₂. This is consistent with the observations that suppressor T cells and macrophages migrate to the renal cortex and medulla during the first 24 hours of obstruction, reaching levels fifteenfold higher than those observed in normal kidneys²⁰⁰ and a parallel rise in TXA₂ release and the fall in GFR.²⁰⁰ These changes can be attenuated by renal irradiation, indicating that obstruction stimulates migration of inflammatory leukocytes, which, in turn, generate vasoconstrictors such as TXA₂.^{200,201} The role of ANG II for this is highlighted because glomeruli isolated from obstructed kidneys showed increased eicosanoid synthesis after ANG II stimulation and that treatment of obstructed animals with converting enzyme inhibitors enhanced GFR and reduced TXA₂ generation by glomeruli isolated from these animals.²⁰² Thus these vasoconstrictors may contribute to regulate R_A and GFR following release of obstruction.

Because vasoconstriction is less severe in animals with bilateral ureteral obstruction, as noted earlier, it is likely that extrarenal factors play a major role in modulating the hemodynamic response of the kidney to obstruction and release of obstruction. In addition to renorenal reflexes already mentioned, various other factors, including accumulation

of volume and solutes such as urea, ANP and its congeners, and other natriuretic substances, may ameliorate the vasoconstrictive effects of obstruction when both ureters are ligated.^{203,204} Following 24 hours of obstruction, GFR is preserved to some degree if the contralateral kidney is also obstructed or removed.²⁰³ In addition, in animals following release of 24 hours of unilateral obstruction, if the urea, salt, and water content of the urine from the contralateral kidney is reinfused into the animal, a striking increase in GFR over standard unilateral obstruction is observed,^{203,205} suggesting that ANP, urea, and other excreted urine solutes have a protective effect and can ameliorate vasoconstriction following release of ureteral obstruction by direct vasodilation of afferent arterioles, constriction of efferent arterioles, and an increase in K_f .

Additional studies in dogs and rats have implicated endothelins as contributors to reduced GFR in obstruction and have suggested that prostaglandin E_2 (PGE₂) and nitric oxide (NO) may play an ameliorating role in glomerular vasoconstriction in the chronic obstructed kidneys.^{206,207} Renal PGE₂ levels increase markedly in obstruction (see later) and in states of extracellular volume expansion, as occurs in bilateral ureteral obstruction. Given the vasodilator effects of PGE₂, it appears likely that increased levels could ameliorate falls in GFR in obstruction. Bilateral obstruction may reduce generation of NO, leading to a net vasoconstrictive effect.²⁰²

PUUO is associated with increased renin–angiotensin–aldosterone system activity, elevated oxidative stress, reduced NO bioavailability, and sensitized afferent arteriolar reactivity and renal autoregulation, leading to blood pressure elevation in both mice and rats.²⁰⁸ Renal denervation in PUUO rats attenuated both hypertension and salt sensitivity, and normalized the renal excretion pattern, whereas the degree of renal fibrosis and inflammation was not changed. This suggests a link between renal nerves, increased blood pressure, and modulation of nicotinamide adenine dinucleotide phosphate oxidase function.

In summary, both intrarenal and extrarenal factors combine to profoundly decrease GFR during and immediately after release of obstruction. The decrease in GFR is caused by a sharp reduction in the number of perfused glomeruli and by a reduction in the SNGFR of functioning nephrons. Decreased K_f and increased R_A reduce SNGFR. Increases in various vasoconstrictors, such as ANG II and TXA₂, as well as other vasoconstrictors, some coming from inflammatory cells, augment these hemodynamic effects. In the setting of bilateral obstruction, retention of urea and other solutes, as well as volume expansion and increases in circulating levels of vasodilators such as ANP, helps offset these vasoconstrictive effects, but only partially.

RECOVERY OF GLOMERULAR FUNCTION AFTER RELIEF OF OBSTRUCTION

The extent of recovery of glomerular filtration following release of obstruction depends on several factors, including the duration and extent of obstruction, the presence or absence of a functioning contralateral kidney, the presence or absence of associated infection, and the level of preobstruction renal blood flow.^{2,209} In a classic experiment in dogs subjected to a 1-week period of complete UUO, GFR fell to 25% of normal on release of the obstruction and recovered gradually to 50% of normal levels 2 years later, indicating

persisting irreversible changes.²¹⁰ In rats, release of UUO of 7- and 14-day duration left residual GFR at 17% and 9% of control levels, respectively, when the contralateral kidney was left in place, and at 31% and 14% when the animals underwent contralateral nephrectomy at the time of release of the obstruction.²¹¹ A similar beneficial effect on the obstructed kidney of contralateral nephrectomy was observed in rats subjected to chronic partial obstruction.²¹¹ As discussed earlier, this beneficial effect likely results from the accumulation of urea and other solutes and increased levels of ANP when the functioning contralateral kidney is absent.

The partial recovery of total GFR following release of obstruction masks a very uneven distribution of blood flow and nephron function. In micropuncture studies, some nephrons never regain filtration function, whereas others reveal striking hyperfiltration.²⁰⁹ It appeared in some studies that surface nephrons exhibited normal SNGFR, whereas the whole-kidney GFR was reduced to 18% of normal.²¹² These results suggest that chronic partial obstruction causes selective damage to juxtamedullary and deep cortical nephrons.^{184,209,212} Similarly, studies of the long-term outcome of complete 24-hour ureteral obstruction revealed that total renal GFR recovered to normal levels by 14 and 60 days after release of obstruction. However, 15% of the glomeruli were not filtering in recovered kidneys, and other nephrons were hyperfiltering. In this model of complete obstruction, there appeared to be no selective advantage for surface glomeruli over deep cortical and juxtamedullary glomeruli.²⁰⁹

Similarly, in the developing kidney, the duration of obstruction and timing of release have a striking impact on long-term renal function. Release after 1 week of obstruction completely prevented development of hydronephrosis, and reduction in renal blood flow and GFR in rats subjected to PUUO at birth,

whereas release after 4 weeks resulted in little or no renal function in the obstructed kidney, demonstrating that early release of neonatal obstruction provides dramatically better protection of renal function than release of obstruction after the maturation process is completed.²¹³ Consistent with this, studies in pigs subjected to neonatal unilateral partial obstruction demonstrated that impaired nephrogenesis resulted in a reduced number of glomeruli in the obstructed kidney.²¹⁴ Preserved whole-kidney function suggests some degree of glomerular hyperfiltration. In line with the hypothesis that hyperfiltration is associated with an increased risk of systemic hypertension,²¹⁵ studies in pigs and rats have demonstrated that renal expressions of neuronal nitric oxide synthase (NOS) and endothelial NOS proteins were lower in animals with hydronephrosis.^{216,217} These findings suggest that the reduced NO response in the obstructed hydronephrotic kidney, and subsequent resetting of the tubuloglomerular feedback (TGF) mechanism, plays an important role in the development of hypertension in hydronephrosis.

EFFECTS OF OBSTRUCTION ON TUBULE FUNCTION

Obstruction severely impairs the ability of renal tubules to transport Na^+ , K^+ , and H^+ , and reduces their ability to concentrate and dilute the urine (Table 37.2).^{2,218–224} The resulting inability to reabsorb water and solutes facilitates postobstructive diuresis and natriuresis. As is the case with glomerular filtration, the extent of disruption of tubular transport depends directly on the duration and severity of the obstruction. Pathologically, prolonged obstruction leads to profound tubular atrophy and chronic interstitial inflammation and fibrosis (see the following section), whereas at early time points following the onset of obstruction, such as at 24 hours, there are only slight structural and ultrastructural changes

Table 37.2 Segmental Reabsorption in Superficial and Juxtamedullary Nephrons and in Collecting Ducts in Normal Rats After Release of Bilateral or Unilateral Obstruction

Site	Normal		After Unilateral Obstruction		After Bilateral Obstruction	
	Water Remaining (%)	Na^+ Remaining (%)	Water Remaining (%)	Na^+ Remaining (%)	Water Remaining (%)	Na^+ Remaining (%)
S ₁	100	100	100	100	100	100
S ₂	44	44	26	26	45	45
S ₃	26	14	21	12	40	22
S ₄	9.4	5	3.2	1.9	25	7
J ₁	100	100	100	100	100	100
J ₂	12	40	42	52	42	62
CD ₁	3.3	2	4.2	3.8	8	6
CD ₂	0.4	0.6	2.9	2.5	16.7	12

In obstruction, increased proportions of filtered salt and water are delivered to the loop of Henle in juxtamedullary nephrons (J₁ and J₂ indicating decreased reabsorption). Delivery of sodium and water to the first accessible portion of the inner medullary collecting duct, labeled CD₁, was also increased, and net sodium and water reabsorption along the inner medullary collecting duct (between CD₁ and CD₂) was diminished in both bilateral and unilateral obstruction. In bilateral obstruction, there was net addition or secretion of sodium and water into the lumen of the inner medullary collecting duct, suggesting that in this setting the inner medullary collecting duct secretes sodium and water. (From Harris KP, Schreiner GF, Klahr S. Effect of leukocyte depletion on the function of the postobstructed kidney in the rat. *Kidney Int.* 1989;36:210–215.)

CD₁, Collecting duct at base of papilla, first accessible portion of inner medullary collecting duct; CD₂, end of collecting duct as it opens into renal pelvis; J_{1–2}, values found in juxtamedullary nephrons; J₁, Bowman space; J₂, tip of loop of Henle; S_{1–4}, values found in superficial nephrons: S₁, Bowman space; S₂, end of the proximal convoluted tubule; S₃, earliest portion of the distal tubule; S₄, end of the distal tubule/beginning of collecting duct.

including mitochondrial swelling, modest blunting of basolateral interdigitations in the thick ascending limb and proximal tubule epithelial cells, as well as flattening of the epithelium and some widening of the intercellular spaces in the collecting ducts.^{2,225,226} The only cell death at early time points is observed at the very tip of the papilla, where focal necrosis may be observed.²²⁵ Because there is so little cell damage, and because of the simplicity of the model, most investigators have examined the effect of 24 hours of complete ureteral obstruction on tubular function. As discussed later, regulation of tubular transport is complex and is due to both direct damage of epithelial cells and the action of extratubular mediators, arising both from the kidney and extrarenal sources.

EFFECTS OF OBSTRUCTION ON TUBULAR SODIUM REABSORPTION

Following release of 24 hours of UUO, volume excretion from the postobstructed kidney is normal or slightly increased^{2,188,203,227} (see Table 37.2). However, as discussed earlier, normal volume excretion occurs in the setting of a markedly reduced (20% of normal) GFR. Consequently, fractional excretion of sodium FE_{Na} is markedly elevated in the postobstructed kidney. After release of bilateral obstruction, salt and water excretion jumps up to five to nine times normal.^{2,188,218,219} Because GFR is also decreased in this setting, FE_{Na} may be twentyfold higher than normal.

The micropuncture studies summarized in Table 37.2 demonstrate that the reabsorption defect following release of obstruction is localized similarly in both unilateral and bilateral ureteral obstruction. Obstruction reduced net salt and water reabsorption in the medullary thick ascending limb (MTAL), the distal convoluted tubule, and the entire length of the collecting duct, including its cortical, outer medullary, and inner medullary segments.²¹⁸

These studies in whole animals were confirmed and extended by a series of studies from multiple laboratories using isolated perfused tubule and cell suspension preparations (Table 37.3). As shown, the segments including proximal straight tubule, MTAL, and cortical collecting duct isolated from unilaterally or bilaterally obstructed animals exhibited

profound impairment of sodium reabsorptive capacity.^{219,220} This finding was confirmed in studies of freshly prepared suspensions of MTAL cells from obstructed kidneys, in which transport-dependent oxygen consumption, a measure of sodium reabsorptive capacity, was markedly reduced.²²¹ Given the major regulatory role of mineralocorticoid in the collecting duct, it is important to note that these decreases in collecting duct sodium reabsorptive capacity occurred in tubules taken from obstructed kidneys, whether or not the animal had been pretreated with mineralocorticoid.^{220,222,223} Because it is highly branched and difficult to perfuse reliably in vitro, transport in the inner medullary collecting duct has been studied in cell suspensions. In these preparations, transport-dependent oxygen consumption was markedly reduced in cells isolated from animals with bilateral obstruction.²²⁴

Taken together, the data derived from micropuncture, tubule perfusion, and cell suspension studies reveal a striking impairment of sodium reabsorption in the proximal straight tubule, the MTAL, and the entire collecting duct. Because these functional derangements occur in the absence of clear-cut ultrastructural damage to the epithelial cells, obstruction likely induces a selective impairment in the regulation of active cellular transport mechanisms. Unlike the situation with glomerular filtration, the functional impairment appears similar in both unilateral and bilateral obstruction.^{220,223,224} Thus it appears that a major component of impaired active transport is likely due to direct tubular cell injury, rather than to the continuous action of natriuretic substances. Added onto this intrinsic injury, natriuretic substances may be responsible for the apparent secretion of sodium and water in the inner medullary collecting duct of animals following release of bilateral obstruction (see Table 37.2).

A combination of studies of cell suspensions and antibody-based targeted proteomics where long-term regulation of renal transporters and channels can be examined in intact animals to understand the integrated response to obstruction has improved the molecular understanding of mechanisms by which tubular epithelial cell sodium reabsorption is impaired in the setting of obstruction. Active tubular Na^+ transport requires an apical entry step (e.g., NKCC2 cotransporter in MTAL or epithelial Na^+ channels [ENaCs] in the

Table 37.3 Function of Isolated Perfused Tubules in Obstructive Nephropathy

	J_v SPCT (nL/mm/min)	J_v PST (nL/mm/min)	ΔCl^- MTAL (mEq/L)	J_v CCT (ADH) (nL/mm/min)
Control	0.75 ± 0.08	0.25 ± 0.02	-37 ± 3	0.90 ± 0.08
Unilateral obstruction	0.73 ± 0.11	0.12 ± 0.03	-9 ± 1	0.22 ± 0.04
Bilateral obstruction	0.80 ± 0.08	0.16 ± 0.02	-10 ± 1	0.23 ± 0.04

The J_v in the SPCT was not affected by obstruction, whereas J_v in PST decreased by 52% in unilateral obstruction (0.12 ± 0.03 vs. 0.25 ± 0.02 nL/mm/min), and similarly in response to bilateral obstruction. In MTAL the ability to lower the perfusate chloride ion concentration was reduced by 76% (-9 ± 1 vs. -37 ± 3 mEq/L), and similarly in response to bilateral ureteral obstruction. Following relief of unilateral obstruction, the ability of the CCT to respond to ADH was reduced by 76% (0.22 ± 0.04 vs. 0.90 ± 0.08 nL/mm/min), and similarly following relief of bilateral obstruction.

ADH, antidiuretic hormone; CCT, cortical collecting tubule; J_v , net fluid reabsorption rate per length of the tubule segment; PST, cortical proximal straight tubule; SPCT, superficial proximal convoluted tubule; ΔCl^- MTAL, change in Cl^- concentration per length of the medullary thick ascending limb.

(Data from Reyes AA, Martin D, Settle S, Klahrs S. EDRF role in renal function and blood pressure of normal rats and rats with obstructive uropathy. *Kidney Int.* 1992;41:403–413.)

collecting duct) coupled to the basolateral Na^+, K^+ -ATPase. In addition, the cell must generate sufficient adenosine triphosphate (ATP) to fuel active transport by the ATPase. Suspensions of MTAL cells from obstructed kidneys exhibited markedly reduced furosemide-sensitive oxygen consumption,²²¹ indicating striking decreases in apical NKCC2 cotransporter activity in these cells. Isotopic bumetanide binding revealed a marked reduction in the number of cotransporter protein molecules available for binding on the membrane, with no change in affinity of binding, indicating that obstruction downregulates the expression of the cotransporter protein on the membrane surface.²²¹ More recent studies using antibody-based targeted approaches clearly showed that obstruction diminishes expression of the cotransporter protein on the MTAL cell apical membrane.²²⁸ Similar approaches demonstrated downregulation of Na^+, K^+ -ATPase of both α - and β -subunits at the transcriptional and posttranscriptional levels.^{228,229}

In the inner medullary collecting duct similar studies demonstrated downregulation of ENaC.²³⁰ Consistent with this, suspensions from obstructed kidneys showed marked decreases in amiloride-sensitive oxygen consumption as well as amiloride-sensitive isotopic sodium entry into hyperpolarized cells.²²⁴

As occurred in MTAL cells, the rates of ouabain-sensitive oxygen consumption and of ouabain-sensitive ATPase were markedly diminished in inner medullary collecting duct cells from obstructed animals, and the levels of both pump subunits were also reduced in these preparations.²²⁴ Patterns of messenger RNA (mRNA) expression were also similar to those in MTAL, indicating transcriptional and posttranscriptional downregulation of pump subunit expression. Using the targeted antibody-based approach demonstrated that in both unilateral and bilateral ureteral obstruction, the expressions of Na^+/H^+ exchanger (NHE₃) and the $\text{Na}^+/\text{PO}_4^{3-}$ exchanger (NaPi-2) were strikingly decreased in the proximal tubule.^{228,231} These changes in sodium transporter expression occurred in both the proximal convoluted and proximal straight tubule, even though the micropuncture and tubule perfusion studies cited earlier revealed preserved proximal convoluted tubule salt reabsorption and inhibition of proximal straight tubule reabsorption.^{231,232} The same studies demonstrated significant downregulation of total transporter protein and apical membrane expression of the distal convoluted tubule Na^+/Cl^- cotransporter, indicating that obstruction likely reduces distal convoluted tubule Na^+ reabsorption by mechanisms similar to those observed in the MTAL and collecting duct.^{228,231}

Taken together, these results demonstrate that obstruction downregulates membrane expression of transporter proteins responsible for apical sodium entry and of basolateral sodium exit. Interestingly, metabolic studies reveal that obstruction reduces activities as well of several enzymes of the oxidative and glycolytic pathways, consistent with a downregulation of metabolic capacity for energy generation in these cells. This may also be enhanced by the observed reductions in the extent of basolateral infolding and in the density of mitochondria in tubules of obstructed kidneys.²²⁵ Interestingly, in MTAL and collecting duct suspensions, obstruction reduces transport-dependent but not transport-independent oxygen consumption, indicating that the rate of ATP generation (oxygen consumption) is not rate-limiting for active transport

in these cells. On this basis, it appears more likely that obstruction-induced reduction of epithelial sodium transport is a regulated process as a result of reduced metabolic demands during obstruction.

The mechanisms and pathways responsible for downregulation of transport proteins in tubular epithelial cells by obstruction remain to a large extent incomplete. Possible signals include the halting of urine flow, increased hydrostatic pressure on tubular epithelial cells, changes in blood flow to the tubules or in interstitial pressure, and generation of natriuretic substances in the kidney that result in long-term inhibition of transporter function. Powerful mass spectrometry analysis of tissue from obstructed rat kidneys and mpkCCD cells has led to proteomic identification of significant changes in more than 100 proteins including those belonging to the cytoskeleton. These findings suggest that obstruction induces acute molecular changes in the renal cytoskeleton, in part, mediated by increased stretch of the renal tubular cells during obstruction.²³³

Obstruction impairs glomerular filtration and urine production is dramatically reduced (stopped in occlusion). Consequently sodium delivery to each tubular segment is reduced and apical membrane Na^+ entry slows dramatically because the electrochemical gradients for Na^+ entry between the stationary apical fluid and the cell interior become increasingly unfavorable for continued sodium transport. Reduced Na^+ entry might then directly stimulate downregulation of transporter activity and expression. In both MTAL and inner medullary collecting duct cells, blocking Na^+ entry by furosemide or amiloride, respectively, promptly reduces ouabain-sensitive oxygen consumption,^{221,234} indicating acute downregulation of Na^+, K^+ -ATPase. In addition, in mineralocorticoid-clamped animals, chronic blockade of Na^+ entry at the MTAL or cortical collecting duct by administration of furosemide or amiloride, respectively, reduced the levels of ouabain-sensitive ATPase in microdissected tubule segments.^{235,236}

These results suggest that the halt in urine flow might represent a major signaling mechanism by which obstruction downregulates Na^+ transport.²³⁴ To test this hypothesis, apical Na^+ entry was inhibited for 24 hours in a cell line that mimics cortical collecting duct cells, A6 cells, grown on permeable supports. When apical Na^+ entry was blocked either by substituting another cation for sodium in the apical solution or by adding amiloride to the apical solution, apical sodium entry was markedly reduced for some hours after the blockade was removed.²³⁷ This downregulation is accompanied by selective reduction in the levels of expression of the β -subunit, but not the α - or γ -subunits, of ENaC in the apical membranes of the A6 cells, but not in whole cell content of these subunits.²³⁸ At the integrated level rats with urinary tract obstruction demonstrated downregulation of α -, β -, and γ -subunits of ENaC, indicating that downregulation of all three subunits may play a role in the impaired sodium reabsorption in obstruction.²³⁰ Interestingly, and in contrast to the results in cell suspensions or whole kidney,^{221,224,228,231} inhibition of apical sodium entry had no effect on expression of either subunit of Na^+, K^+ -ATPase.²³⁸ These results provide direct evidence that reductions in the rate of Na^+ entry, which may occur when urine flow is blocked, can directly downregulate Na^+ transport in renal epithelial cells.

In addition to the direct effects of halting urine flow, changes in intrarenal mediators and subcellular pathways

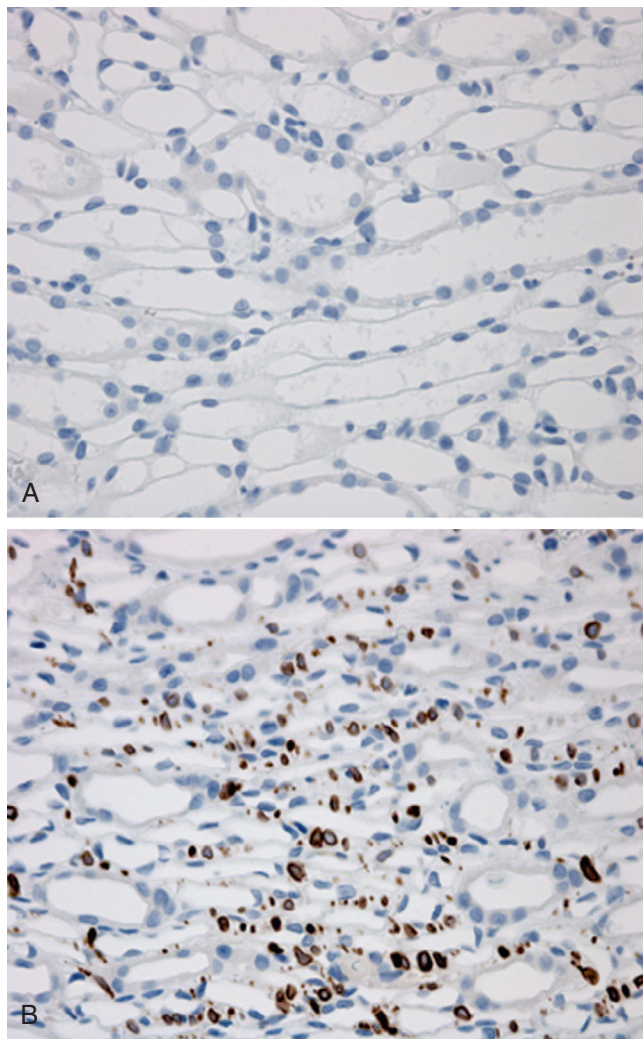


Fig. 37.8 Immunohistochemistry for cyclooxygenase-2 in the inner medulla of the kidney of sham-operated rats (A), and rats subjected to 24 hours of bilateral ureteral obstruction (B). There is strong labeling at the base of the inner medulla in obstructed kidneys located exclusively to the interstitial cells (B), and labeling is not detectable in sham kidneys (A).

likely play a critical role in the reduction of salt transport observed with obstruction. Obstruction markedly accelerates the already rapid generation of PGE_2 in the renal medulla.^{195,196,199,239} The molecular basis for this is a dramatic medullary cyclooxygenase-2 (COX-2) induction (Fig. 37.8).^{239,240} Consistent with the known effect of PGE_2 to markedly inhibit Na^+ reabsorption in the MTAL, as well as in the cortical and inner medullary collecting ducts,^{241–243} COX-2 inhibition in rats with obstruction and release of obstruction attenuated the downregulation of NHE_2 , NKCC2 , and Na^+, K^+ -ATPase.^{240,244} From these results, obstruction likely reduces apically localized sodium cotransport proteins in the tubule epithelium and sodium pump activity in tubular epithelia in part by increasing renal levels of PGE_2 .

As discussed earlier, obstruction brings on a mononuclear infiltrate in the kidney²⁰⁰; and this infiltrate tends to follow a peritubular distribution.²⁰⁰ When obstructed kidneys were irradiated, the level of medullary inflammation was diminished, and there was a modest decrease in the fractional excretion of sodium.²⁰¹ In addition, it has been shown that

obstruction causes an enhanced renal ANG II generation. This may have important implications for regulation of renal sodium handling. Blockade of the ANG II receptor (AT1) was associated with a marked attenuation of downregulation of NHE_3 and NKCC2 which was paralleled by a reduction in renal sodium loss.⁵⁷

In summary, obstruction reduces net reabsorption of sodium in several nephron segments, including the proximal straight tubule, the MTAL, and the cortical and inner medullary collecting ducts, by downregulating the expression and activities of specific transporter proteins. Several signals mediate this downregulation, including the cessation of urine flow with its attendant reduction of the rate of Na^+ entry across the apical membrane, increased levels of natriuretic substances such as PGE_2 , and infiltration of the obstructed kidney by mononuclear cells.

When both ureters are obstructed, extrarenal factors markedly enhance the sodium-wasting tendency already present in the obstructed kidney. One mechanism involves the volume expansion that occurs when bilateral obstruction ablates all renal function. Volume expansion impairs activity in the sympathetic nervous system, reduces circulating levels of aldosterone, and, along with reduced renal clearance, increases levels of ANP. Reduced sympathetic tone and aldosterone, coupled with increased ANP, markedly stimulate sodium excretion. ANP likely represents a particularly important mediator of salt wasting in bilateral obstruction. Levels of ANP are markedly elevated in bilateral, but not in unilateral obstruction.²⁴⁵ ANP enhances salt wasting at several nephron segments. By blocking renin release in the macula densa and angiotensin action in the proximal tubule, ANP reduces proximal tubule sodium reabsorption.^{204,245,246} ANP also reduces aldosterone release and directly inhibits sodium reabsorption in the collecting ducts.^{204,245,246} In agreement with this mechanism, infusion of ANP into animals in which obstruction has just been released leads to marked increases in sodium and water excretion.²⁴⁵ Moreover, efforts to reduce circulating ANP levels following bilateral obstruction attenuated sodium excretion somewhat.²⁴⁵

In addition, accumulation of urea and other solutes enhance sodium wasting by obstructed kidneys. Following release of 24 hours of unilateral obstruction, removal or obstruction of the contralateral kidney markedly enhances salt wasting by the postobstructed kidney.²⁰³ If the contralateral kidney is left in place but amounts of urea, sodium, and water equivalent to what the contralateral kidney is excreting are infused into the animal, there is a striking increase in sodium excretion in both the obstructed and the contralateral kidney.^{203,205} On this basis, bilateral obstruction induces hormonal changes and promotes accumulation of solutes and volume that together enhance natriuresis from the obstructed kidney.

EFFECTS OF OBSTRUCTION ON URINARY CONCENTRATION AND DILUTION

Because obstruction eliminates the ability of the renal tubules to concentrate and dilute the urine, urine osmolality following release of obstruction in humans and experimental animals approaches that of plasma.^{2,247,248} Dilution of the urine requires that the thick ascending limb reabsorbs sodium without water and that the collecting duct maintain the dilute urine by not reabsorbing water along its length, despite the presence

of a concentrated medullary interstitium.²⁴⁹ Concentration of the urine requires active sodium reabsorption in the thick limb and the action of the countercurrent multiplier to generate a concentrated medullary interstitium, as well as the ability of the collecting duct to insert the vasopressin-regulated water channel aquaporin-2 (AQP2) into the apical membrane.^{250,251}

Obstructive nephropathy disrupts several of these mechanisms.^{231,247,248,252} As noted earlier, obstruction also markedly reduces MTAL sodium reabsorption, limiting this segment's ability to dilute the urine and to generate a high medullary interstitial osmolality. Indeed, interstitial osmolality has been shown to be reduced in obstructed kidneys.² In addition, collecting ducts isolated from obstructed kidneys reveal normal basal water permeabilities, but a marked reduction in their ability to increase water permeability in response to antidiuretic hormone or other stimulants of cyclic adenosine monophosphate (cAMP) accumulation in the cells. As was the case with sodium transport, the effects were similar in unilateral and bilateral obstruction.^{247,252} Detailed mechanistic studies show that obstruction markedly reduces transcription of mRNA encoding AQP2, as well as synthesis of AQP2 protein, and that collecting duct cells in obstructed kidneys do not traffic AQP2-containing vesicles effectively to the apical surface in response to vasopressin or increased cAMP.^{247,251–253} Part of this failure in trafficking results from a decrease in phosphorylation of AQP2 in obstructed kidneys²³¹ and likely also due to the fact that V2 receptor protein expression is downregulated.²⁵⁴ Redistribution of AQP2 and AQP2 phosphorylated at ser261 to more intracellular localizations after bilateral obstruction and colocalization with the early endosomal marker EEA1 and the lysosomal marker cathepsin D suggest that early downregulation of AQP2 could in part be caused by degradation of AQP2 through a lysosomal degradation pathway.²⁵⁵ In addition, UO markedly decreases synthesis and deployment to the basolateral membrane of aquaporin-3 and -4; when AQP2 is in the apical membrane, these aquaporins mediate the water flux across the basolateral membrane.²³¹ Enhancing the causal relationship of the changes in aquaporin activity and ability to concentrate the urine, expression of AQP2 remains suppressed for 7 days following relief of the obstruction, and the rise in urinary concentration parallels the recovery in AQP2 expression.^{231,247,248,252} The fact that collecting ducts from obstructed kidneys do not respond to cAMP indicates that the lesion also involves sites beyond the receptor for antidiuretic hormone.²⁵⁴

Consistent with the hypothesis that PGE₂-mediated inhibition of collecting duct water permeability does not directly affect cAMP levels but may have post-cAMP effects rather than actions via cAMP regulation,²⁵⁶ recent experiments have shown that COX-2 inhibition prevented dysregulation of AQP2 in obstructed kidneys where COX-2 protein expression was markedly increased (Fig. 37.8).^{240,257}

On the basis of these results it can be concluded that the defect in urinary dilution in obstruction is due to reduced ability of the thick ascending limb to dilute the urine by transporting salt from the lumen of the tubule to its basolateral side. The collecting duct in obstructed kidneys maintains its low water permeability in the absence of antidiuretic hormone, so that the failure to dilute the urine is not due to collapse of osmotic gradients in the collecting duct. The inability to concentrate the urine results from the failure of the thick

limb to generate a concentrated interstitium, as well as the inability of the collecting duct to synthesize and to traffic AQP2 and other water channels in response to antidiuretic hormone.

EFFECTS OF RELIEF OF OBSTRUCTION ON URINARY ACIDIFICATION

Obstruction dramatically reduces urinary acidification in both experimental animals and humans. In humans, release of obstruction does not lead to bicarbonate wasting, indicating that proximal tubular bicarbonate reclamation is maintained. By contrast, in both experimental animals and patients following release of obstruction, the urine pH does not decrease in response to an acid load, indicating that obstruction impairs the ability of the distal nephron to acidify the urine.^{258–260} This defect likely involves proton transport proteins both in the collecting duct^{258,259} and in the proximal tubule and thick ascending limb of Henle.^{260,261}

Reduced collecting duct acid secretion could result from defects in H⁺ (H⁺, ATPase or H⁺ K⁺-ATPase) or HCO₃⁻ (e.g., Cl⁻/HCO₃⁻ exchange) transport pathways, backleak of protons down their electrochemical gradient from the lumen to the basolateral side of the tubule, or, in the cortical collecting duct, the failure to generate a sufficiently lumen-negative transepithelial voltage.^{258,259,262} As described in detail earlier, obstruction reduces the activity of apical ENaC in the cortical collecting duct; the resulting loss of luminal negativity may attenuate acid secretion in these segments.^{258,259}

In the rat inner medullary collecting duct (studied by micropuncture) and in isolated perfused rat and rabbit outer medullary collecting duct, obstruction markedly reduces luminal acidification rates.²⁵⁸ Because Na⁺ transport does not play a major role in acidification in these segments, the defect must be due to direct inhibition of acid or HCO₃⁻ transport pathways, or backleak of protons from lumen to interstitium.²⁶² At low perfusion rates, outer medullary collecting ducts from obstructed animals maintain the ability to generate steep pH gradients,²⁵⁸ indicating that obstruction does not block the ability of the tubule to prevent back flux of protons. By contrast, at high perfusion rates, acidification was markedly lower in tubules from obstructed, as opposed to normal, kidneys,²⁵⁸ demonstrating that obstruction inhibits activity or expression of H⁺ or HCO₃⁻ transport pathways.

Antibody-based targeted studies examining the Cl⁻/HCO₃⁻ exchanger and subunits of the H⁺, ATPase revealed reduced expression of these transporters in collecting ducts of unilaterally obstructed compared with contralateral, and control kidneys.^{260,262} Two possible mechanisms of reduced acid secretion were explored.²⁶² One was that the intercalated cells in obstructed kidneys would exhibit a high proportion of “reverse” orientation, with the proton pump in the basolateral membrane, and the Cl⁻/HCO₃⁻ exchanger in the apical membrane. The other possibility was that the orientation of intercalated cells would not change, but there would be reduced expression of the H⁺ or HCO₃⁻ transporter. The orientation of the intercalated cells was not altered by obstruction. However, obstruction did reduce the appearance of H-ATPase along the apical membranes of intercalated cells, without altering the total content of H⁺-ATPase in extracts of renal cortex or medulla, in unilaterally obstructed, as compared with contralateral kidneys.²⁶² In obstructed kidneys, fewer intercalated cells exhibited an apical labeling pattern

and many that did showed discontinuities or gaps in apical membrane labeling,²⁶² suggesting that obstruction inhibits trafficking of H⁺-ATPase to the apical membranes of intercalated cells. However, this disorder alone cannot account for the entire acidification defect in obstructive nephropathy, because the labeling pattern returns to control levels as the obstruction persists, while the acidification defect remains.²⁶² In addition, the extent of the decrease in labeling appears to be too small to account for the profound defect in acidification.

In addition to defective collecting duct, H⁺ transport–reduced generation of the main buffer that carries acid equivalents in the urine, ammonia has also been observed in kidneys released from obstruction. Cortical slices of obstructed kidneys exhibit reduced glutamine uptake and oxidation, reduced gluconeogenesis, and reduced total oxygen consumption, all adding up to a reduced ability to generate ammonia from glutamine.^{263,264}

EFFECTS OF RELIEF OF OBSTRUCTION ON EXCRETION OF POTASSIUM

As with sodium excretion, potassium excretion increases markedly following release of bilateral obstruction.^{265,266} Micropuncture and microcatheterization studies show that proximal potassium reabsorption is unchanged by obstruction while potassium is more rapidly secreted in the collecting duct, likely due to increased distal delivery and therefore more rapid distal luminal flux of sodium and volume following release of obstruction.^{218,265} By contrast, following release of unilateral obstruction, potassium excretion falls roughly in proportion to the reduction in GFR,²⁶⁷ an effect that may be related to reduced distal delivery of sodium. However, administration of sodium sulfate in this state does not stimulate potassium excretion in obstructed kidneys as it does in controls, suggesting that collecting ducts in unilateral obstructed kidneys have an intrinsic defect in potassium secretion.²⁶⁸ This intrinsic defect may represent a response similar to the downregulation of sodium transporters in obstructed kidneys described in detail earlier. The kaliuretic effect observed in bilateral obstruction may well be due as well to the influence of elevated levels of ANP, which, at high levels, can stimulate potassium secretion in the distal nephron.

EFFECTS OF RELIEF OF OBSTRUCTION ON EXCRETION OF PHOSPHATE AND DIVALENT CATIONS

When bilateral ureteral obstruction is released, phosphate excretion rises in proportion to sodium excretion.^{228,231,269,270} Phosphate restriction before the release of the obstruction prevents phosphate accumulation during bilateral obstruction, thereby blocking the increase in phosphate excretion.²⁶⁹ This can also be achieved by blockade of ANG II–mediated effects, highlighting the importance of enhanced renal ANG II levels in the obstructed kidney.⁵⁷ In addition, phosphate wasting of similar magnitude to that observed following release of bilateral obstruction can be duplicated by phosphate loading of normal animals.²⁶⁹ By contrast, release of unilateral obstruction results in phosphate retention, likely due to reduced GFR and avid proximal phosphate reabsorption.²⁷¹ Calcium excretion may be increased or decreased, depending on whether the obstruction is unilateral or bilateral, and depending on the species studied.^{269,271} Magnesium excretion is markedly increased following release of either bilateral or

unilateral obstruction. This magnesium wasting probably occurs because both forms of obstruction markedly attenuate thick ascending limb sodium reabsorption, leading to reduced positive luminal transepithelial voltages and therefore a reduced driving force for lumen-to-basolateral magnesium flux across the paracellular pathway.²⁷²

PATHOPHYSIOLOGY OF RECOVERY OF TUBULAR EPITHELIAL CELLS FROM OBSTRUCTION OR OF TUBULOINTERSTITIAL FIBROSIS

An important focus of many experimental studies in obstructive nephropathy has been devoted to the renal effects of longer-term obstruction.²⁷³ In part these studies use UO as a convenient model for chronic renal damage, because the timing of the injury is clear and because the extent of injury should be reproducible from animal to animal.²⁷³ These studies, which have been conducted almost entirely in rodents, have elucidated an overall pathway for renal tubular epithelial damage, and have identified several potential targets for intervention. Obstruction inhibits oxidative metabolism and promotes anaerobic respiration, leading to decreased ATP levels and increased levels of adenosine diphosphate (ADP) and AMP.^{264,274,275} In addition, obstruction alters a wide variety of metabolic enzymes, as well as the expression of many different gene products.^{264,275–277} These changes are summarized in [Box 37.3](#). Many of these changes are difficult to link mechanistically with changes in GFR or tubular transport function observed in obstruction. It is possible, however, that reduced ability to generate ATP, along with reductions in Na⁺,K⁺-ATPase expression, contributes to the natriuresis observed following release of obstruction (see earlier discussion).

It is thought that chronic obstruction damages tubular epithelial cells by increasing hydrostatic pressure, reducing blood flow (due to the renal vasoconstriction that occurs in obstruction, see earlier discussion), and increasing oxidative stress.²⁷³ In response, tubular epithelial cells release a number of autocrine factors and cytokines, including ANG II,^{273,278} transforming growth factor-beta (TGF-β),^{278,279} platelet activator inhibitor,²⁸⁰ and tumor necrosis factor (TNF).²⁸¹ These factors, along with the presence and increase in levels of adhesion factors, lead to the infiltration of the renal interstitium with inflammatory cells, including macrophages. These in turn release additional cytokines. All these factors accelerate the development of interstitial fibrosis by increased extracellular matrix, cell infiltration, apoptosis, and accumulation of activated myofibroblasts.²⁸² In addition, there is upregulation of various receptors that are targeted by autocrine factors and cytokines including both type 1 and type 2 angiotensin receptors (AT₁R and AT₂R).²⁸³ The complex cellular migration process may become amplified by epithelial–mesenchymal transition (EMT),²⁸⁴ which is characterized by downregulation of epithelial marker proteins such as E-cadherin, zonula occludens-1, and cytokeratin; loss of cell-to-cell adhesion; upregulation of mesenchymal markers including vimentin, α-smooth muscle actin, and fibroblast-specific protein-1; basement membrane degradation; and migration to the interstitial compartment.²⁸⁴ The entire cascade leads to tubulointerstitial fibrosis and permanent loss of renal function,

Box 37.3 Effects of Urinary Tract Obstruction on Renal Enzymes and Renal Gene Expression

Changes in Energy and Substrate Metabolism

Decreased oxygen consumption
Decreased substrate uptake
Increased anaerobic glycolysis
Decreased ATP/(ADP + AMP)
Decreased ammoniogenesis

Changes in Enzyme Activity

Decreased
Alkaline phosphatase
Na⁺, K⁺-ATPase
Glucose-6-phosphatase
Succinate dehydrogenase
NADH/NADHP dehydrogenase
Increased
Glucose-6-phosphate dehydrogenase
Phosphogluconate dehydrogenase
Mitogen-activated protein kinases
Matrix metalloproteinase-2/-9
Mast cell protease 1 (chymase)

Changes in Gene Expression

Reduction in glomerular G_{as} and G_{α₁₁} proteins
Reduction in preproepidermal growth factor and Tamm-Horsfall protein
Transient induction of growth factors FOS and MYC
Striking induction of cellular damage (TRPM2) genes
Induction of plasminogen activator gene

ADP, Adenosine diphosphate; AMP, adenosine monophosphate; ATP, adenosine triphosphate; NADH, nicotinamide adenine dinucleotide (reduced).

which may continue to progress after the obstruction has been relieved (Fig. 37.9).

It has been hypothesized that changes in the intratubular dynamic forces—so-called tubular stretch—in urinary tract obstruction also are an important determinant for development of tubulointerstitial fibrosis in the kidney.²⁸⁵ Thus both in vivo²⁷⁹ and in vitro models^{286,287} of obstructive uropathy demonstrate that tubular stretch induces robust expression of TGF-β1, activation of tubular apoptosis, and induction of nuclear factor-κB signaling, which contribute to the inflammatory and fibrotic milieu.^{285,288} Because fibrosis is absent in mast cell-deficient mice with UUO, this suggests that mast cells play an important role in induction of the inflammation and development of tubulointerstitial fibrosis in obstruction.²⁸⁹ Furthermore, the importance of the renin-angiotensin system for the development fibrosis was also underscored by the observation that mast cells release renin possibly stimulated by autocrine histamine release.²⁸⁹ Interaction between DNA repair and parenchymal inflammation of the kidney may also play an important role for regulating fibrosis development during ureter obstruction. Apurinic/aprimidinic endonuclease (APE1) is a multifunctional protein with important functions in genome maintenance and the regulation of multiple transcription factors. The base excision repair pathway is the major pathway for repairing oxidative DNA base lesions. These lesions are formed when reactive

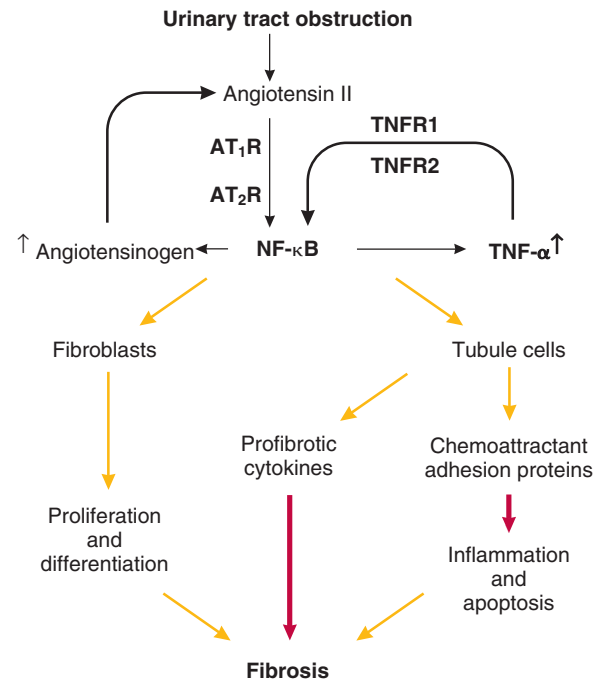


Fig. 37.9 Urinary tract obstruction causes an enhanced expression of angiotensin II (ANG II). The regulation of gene expression by ANG II occurs through specific receptors that are ultimately linked to changes in the activity of transcription factors within the nucleus of target cells. In particular, members of the nuclear factor (NF)-κB family of transcription factors are activated, which, in turn, fuels at least two autocrine-reinforcing loops that amplify ANG II and tumor necrosis factor (TNF)-α formation. AT₁R, Type 1 angiotensin receptor; AT₂R, type 2 angiotensin receptor; TNFR1, TNF-α receptor 1; TNFR2, TNF-α receptor 2.

oxygen species attack DNA²⁹⁰ and APE1 plays a central role in base excision repair and is estimated to be responsible for more than 95% of abasic site incision activity in human cells.²⁹¹ APE1 levels were increased in almost all zones of the kidney after 7 days of unilateral ureter obstruction and the transcription regulatory activity of APE1 is involved in the regulation of the profibrotic factor connective tissue growth factor,²⁹² suggesting that APE1 is a protein with a potential complex role for development of tubulointerstitial fibrosis and based on dual functions: DNA repair and transcription regulation.

Thus the pathogenesis of renal fibrosis is a progressive and complicated process involving multiple molecular pathways and cellular targets including ANG II as highlighted already.^{4,190} ANG II is known to stimulate production of TGF-β, which is critical for the development tubulointerstitial fibrosis.²⁹³ This has been highlighted by the recent in vivo documentation of a relationship between ANG II and TGF-β in genetically defined mouse models of Marfan syndrome.²⁹⁴ Interestingly, recent studies show that selected manifestations of Marfan syndrome reflect excessive signaling by TGF-β.²⁹⁵ Moreover, systemic antagonism of TGF-β through administration of a TGF-β neutralizing antibody or treatment with losartan prevented aortic aneurysm²⁹⁴ and normalized muscle architecture, repair, and function in Marfan-like fibrillin-1-deficient mice.²⁹⁵ Collectively these studies propose that mast cells' release of renin and local ANG II formation in the obstructed kidney may lead to both vasoconstriction, which causes reduction in renal blood flow as well as GFR, and fibroblast and macrophage

activation, which increases TGF- β levels and causes fibrosis of the kidney. Interestingly recent data suggest that mast cells also have the capacity to release chymase, a protease, which may limit development of tubulointerstitial fibrosis by decreasing infiltration of inflammatory cells and release of proinflammatory and profibrotic chemokines and cytokines.²⁹⁶

Injection of COX-2 chitosan/siRNA (small interfering RNA) nanoparticles in mice subjected to 3-day UUO diminished the obstruction-induced COX-2 expression. Likewise, macrophages in the obstructed kidney had reduced COX-2 immunoreactivity, and histological examination showed lesser tubular damage in COX-2 siRNA-treated ureter-obstructed mice.²⁹⁷ Parenchymal inflammation, assessed by TNF- α and interleukin-6 (IL-6) mRNA expression, was attenuated by COX-2 siRNA. Furthermore, treatment with COX-2 siRNA reduced heme oxygenase-1 and cleaved caspase-3 in mice with ureter obstruction, indicating lesser oxidative stress and apoptosis. Thus treatment with chitosan/siRNA nanoparticles to knock down COX-2 accumulated in macrophages demonstrates a novel strategy to prevent ureter obstruction-induced kidney damage.

Several studies have shown that antagonism of ANG II, TGF- β , TNF, or factors that attract inflammatory cells may ameliorate postobstructive renal damage.^{273,278–282,298–300} Similarly, augmentation of expression of factors that favor epithelial growth and differentiation, such as hepatocyte growth factor,³⁰¹ insulin-like growth factor, or BMP-7,²⁷³ may also have a protective effect.

From multiple studies, it has been suggested that the main mechanism that is responsible for the onset of the pathophysiological cascades is the increased pressure in the renal pelvis, which leads to increased pressure in the parenchyma, and subsequently mechanical stress, which leads to activation of stretch and swelling-activated cation channels within focal adhesions of the epithelial cells causing subsequent influx of Ca²⁺.^{287,288} This causes oxidative stress and stimulates migration of macrophages to the obstructed kidney. However, the complexity of the process and the experience from multiple studies demonstrating that there is no single pathway responsible for the cellular changes were highlighted by the potential role of infiltrating macrophages in the pathophysiology of inflammation during UUO. It was demonstrated that activation of AT₁R on macrophages in UUO is prerequisite for suppression of their release of the proinflammatory cytokine IL-1.³⁰² This indicates that a key role of AT₁R on macrophages is to protect the kidney from fibrosis by limiting activation of IL-1 receptors in the obstructed kidney. Further, this finding may show implications for the design of novel potent therapies to overcome the shortcomings of global angiotensin receptor blockade. Thus the process leading to kidney fibrosis is complex, and numerous processes contribute to regulate the cellular responses responsible for these pathophysiological changes.

Given species differences and the fact that obstruction in humans is often partial, the animal models may not predict entirely the behavior of postobstructive kidneys in humans. However, if the studies are relevant to human obstructive nephropathy, they suggest that patients undergoing release of obstruction may benefit from therapies that block proapoptotic, proinflammatory, or profibrotic mediators, or from treatments that stimulate epithelial cell growth and differentiation.^{273,278–282,298–300}

Experimentally, protection from obstruction-induced detrimental effects on renal function can also be achieved by NO supplementation. This can either be accomplished by angiotensin-converting enzyme inhibition, which increases kinin levels and subsequently increases NO formation, or by stimulation of endogenous NO synthase (NOS) with L-arginine.³⁰³ L-Arginine is a semiessential amino acid and is also substrate and the main source for generation of NO via NOS. Importantly, chronic unilateral obstruction in mice leads to significant reduction in inducible (i)NOS activity, and the obstructed kidney of iNOS knockout mice exhibited significantly more apoptotic renal tubules than controls, underscoring the important role NO plays for protecting the cellular functions in the obstructed kidney.³⁰⁴ Dietary L-arginine supplementation attenuated renal damage of a 3-day UUO in rats, indicating that L-arginine treatment may be a pharmacologically useful avenue in obstructive nephropathy.²⁷³ It was also shown that several of the detrimental effects of obstruction can also be attenuated by treatment with α -melanocyte-stimulating hormone (α -MSH), which is a potent antiinflammatory hormone, supporting the view that inflammation is a crucial determinant for the onset of renal deterioration in urinary tract obstruction.³⁰⁵ Interestingly, it was recently demonstrated that recombinant human erythropoietin (rhEPO) treatment inhibits the progression of renal fibrosis in the obstructed kidney and attenuates the TGF- β 1-induced EMT, suggesting that the renoprotective effects of rhEPO could be mediated, at least partly, by inhibition of TGF- β 1-induced EMT.³⁰⁶

During renal fibrosis, myofibroblasts accumulate in the interstitium of the kidney, leading to deposition of extracellular matrix and organ dysfunction, and EMT may play an important role in this. It was shown that the reactivation of *Snail* (encoding snail family zinc finger 1, known as Snail1) in mouse renal epithelial cells is required for the development of fibrosis in the kidney. Damage-mediated Snail1 reactivation induces a partial EMT in tubular epithelial cells that, without directly contributing to the myofibroblast population, relays signals to the interstitium to promote myofibroblast differentiation and fibrogenesis and to sustain inflammation.³⁰⁷ Importantly, it was shown that Snail1-induced fibrosis can be reversed in vivo and that obstructive nephropathy can be therapeutically ameliorated in mice by targeting Snail1 expression. These results reconcile conflicting data on the role of the EMT in renal fibrosis and provide avenues for the design of novel antifibrotic therapies.³⁰⁷ The immune infiltrate in mouse kidneys with fibrotic obstructive nephropathy was recently characterized and demonstrated that lymphocytes and macrophages were the most abundant immune cells present in the fibrotic kidneys.³⁰⁸ Interestingly, preventing tubular epithelial cells from undergoing EMT also decreased immune infiltration, suggesting that crosstalk between tubular epithelial cells and immune cells is of fundamental importance in sustaining inflammation and fibrosis.

Several other therapeutic avenues for direct targeting of specific pathways involved in the fibrotic process of obstructive nephropathy have been demonstrated. Apoptosis signal-regulating kinase 1 (ASK1) acts as an upstream regulator for the activation of p38 mitogen-activated protein kinase (MAPK) and c-Jun N-terminal kinase (JNK) in kidney disease. Mice lacking the Ask1 gene are healthy with normal homeostatic functions and are protected from renal interstitial

fibrosis induced by ureteric obstruction.³⁰⁹ Thus the future may show therapeutic potential for ASK1 inhibitors to treat renal interstitial fibrosis.

FETAL URINARY TRACT OBSTRUCTION

Obstructive uropathy comprises the largest fraction of identifiable causes of renal insufficiency and renal failure in infants and children. Compared with adult obstructive nephropathy, fetal obstructive nephropathy is particularly devastating because renal growth and continued nephron development are impaired by the progression of fibrosis. Several studies have examined aspects of obstructive nephropathy in the newborn using a neonatal rat model of unilateral obstruction and the pathophysiology involved in fetal urinary tract obstruction will be discussed in Chapter 72. Briefly, fetal urinary obstruction may lead to changes in tissue differentiation. At the time of birth, the rodent kidney is not fully developed and is representative of human renal development at about the mid-trimester. Animal models reveal that fetal obstruction causes aberrations of morphogenesis, gene expression, cell turnover, and urine composition.^{310,311} The earlier the kidney is obstructed in utero, the greater will be the changes in renal tissue.^{310,311} After birth, obstruction may affect renal growth, especially in neonates and during the first year of life, but the obstruction will not cause tissue dedifferentiation.

Studies have demonstrated the upregulation of the renin-angiotensin system, as well as involvement of other substrates (TGF- β 1, endothelin-1, and many other mediators) in obstructed kidneys.³¹⁰⁻³¹³ The exact mechanisms of action of these molecules in the alteration of renal morphogenesis are not fully understood. It is not well known either if obstruction alone is enough to induce renal dysplasia^{310,311} or if the latter results from secondary obstruction-induced mesenchymal disruption. To know the exact role of obstruction in the kidney malformation is very important clinically, because, as mentioned earlier, it is now possible to detect and potentially relieve obstruction in utero. The critical role of AT₂R in the process of ureteric bud in kidney development has been highlighted.³¹⁴ In this regard polymorphisms in the AT₂R gene have been shown to be associated with UPJ obstruction.³¹⁵ If urinary obstruction is not the cause of subsequent renal impairment, then some may argue whether it is worthwhile to relieve the obstruction in utero. However, in experimental models, obstruction in utero can cause pulmonary hyperplasia and renal impairment directly or indirectly, leading to significant morbidity and mortality.^{310,311,316} In addition, shunting of urinary outflow from obstructed kidneys in animals before the end of nephrogenesis may allow reversal of the arrest of glomerulogenesis seen in this setting,³¹⁷ favoring early intervention.³¹⁶⁻³¹⁹ The changes in renal gene expression and protein production afford many potential biomarkers of disease progression and targets for therapeutic manipulation.³²⁰

TREATMENT OF URINARY TRACT OBSTRUCTION AND RECOVERY OF RENAL FUNCTION

Once the presence of obstruction is established, intervention is usually strongly indicated to relieve it. The type of intervention depends on the location of the obstruction, its degree,

and its etiology, as well as the presence or absence of concomitant diseases and complications and the general condition of the patient.³²¹ As detailed later, treatment options for obstruction include a wide spectrum of approaches, from active surveillance or minimally invasive endourologic techniques to open, laparoscopic, or robotic pyeloplasty. The main goal of therapy is to relieve symptoms and maintain or improve renal function, but it remains difficult to define treatment success after therapy. The initial emphasis focuses on prompt relief of the obstruction, followed by the definitive treatment of its cause. Obstruction below the bladder (e.g., benign prostatic hyperplasia or urethral stricture) is easily relieved with placement of a urethral catheter. If the urethra is impassable, suprapubic cystostomy may be needed. For obstruction above the bladder, insertion of a nephrostomy tube or ureteral stent may be indicated. The urgency of the intervention depends on the degree of renal function, the presence or absence of infection, and the overall risk of the procedure.³¹⁰ The presence of the infection in an obstructed urinary tract, or urosepsis, represents a urologic emergency that requires immediate relief of the obstruction, in addition to antibiotic treatment. Acute kidney injury, associated with bilateral ureteral obstruction or with the obstruction of single functioning kidney, also calls for emergent intervention.

Calculi, the most common form of acute unilateral urinary obstruction, can usually be managed conservatively with analgesics for control of pain and intravenous fluids to increase urine flow. Approximately 90% of stones smaller than 5 mm pass spontaneously, but as they get larger, spontaneous stone passage becomes progressively less probable. Active efforts to fragment or remove the stone are indicated for persistent obstruction, uncontrollable pain, or urinary tract infection. Current possibilities for treatment include extracorporeal shock wave lithotripsy (which may require ureteral stent placement if the patient is symptomatic),³²² ureteroscopy with stone fragmentation (usually with laser lithotripsy), and, in rare cases, open excision of the stone.^{172,173,323} In general, a combination of lithotripsy and endourological procedures will succeed in removing the stone. In the past, complex stones high up in the ureter or in the renal pelvis have been difficult to remove without open surgery. However, improved methods of lithotripsy, including the use of laser lithotripsy through the ureteroscope, have made more stones amenable to fragmentation, while miniaturization of flexible ureteroscopes has made the entire upper urinary tract accessible in nearly all patients, except those with severe anatomic abnormalities.^{172,173} Once the stone has been removed, of course, appropriate medical therapy is needed to prevent recurrence.¹⁷³

Intramural or extrinsic ureteral obstruction may be relieved by placement of a ureteral stent through the cystoscope.³²² If this cannot be accomplished or is ineffective (especially in cases of extrinsic ureteral compression by tumor), then nephrostomy tubes will need to be inserted to effect prompt relief of the obstruction.³²²

For infravesical obstruction due to benign prostatic hyperplasia, surgery can be safely delayed or completely avoided in patients with minimal symptoms, lack of infection, and an anatomically normal upper urinary tract.³²⁴ If needed, transurethral resection of the prostate, laser ablation, or other techniques can be used for definitive treatment. Internal urethrotomy with direct visualization may be effective in the treatment of urethral strictures, as dilation usually has only

temporary effect. Suprapubic cystostomy may be necessary in patients with impassable urethral strictures, followed by open urethroplasty to restore urinary tract continuity, when possible.

Patients with neurogenic bladder require a variety of approaches, including frequent voiding, often by external compression or Credé method, medications to stimulate bladder activity or relax the urethral sphincter, and intermittent catheterization using meticulous techniques to avoid infection.^{44,325} Long-term indwelling bladder catheters should be avoided because they increase the risk of infection and renal damage. If more conservative measures such as frequent voiding or intermittent catheterization are not effective, ileovesicostomy or other forms of urinary diversion should be considered. Electrical stimulation has also been attempted with varying success.³²⁶

In many forms of obstruction, initial stabilization of the patient's condition is followed by a decision as to whether to continue observation or to move on to definitive surgery or nephrectomy. The actual course chosen depends on the likelihood that renal function will improve with the relief of obstruction. Factors that help decide whether to operate and what form of surgical intervention to use include the age and general condition of the patient, the appearance and function of the obstructed kidney and the contralateral one, the cause of the obstruction, and the absence or presence of infection.³²⁷ As noted earlier, the extent of recovery of renal function depends on the extent and duration of the obstruction.

Robotic surgery has evolved from simple extirpative surgery to complex reconstructions including pyeloplasty for hydronephrosis, which is feasible and safe.^{328,329} Interestingly, a recent metaanalysis comparing robotic, laparoscopic, and open pyeloplasty in children showed that robotic pyeloplasty might offer shorter hospital stays, lower analgesia requirements, and lower estimated blood loss. The postoperative success rate was comparable in the two groups, but there was a significantly higher complication rate and higher costs in the robotic group.³³⁰ A detailed discussion of the indications and surgical techniques for intervention to treat urinary tract obstruction is beyond the scope of this chapter and may be found in other sources.^{321,331}

ESTIMATING RENAL DAMAGE AND POTENTIAL FOR RECOVERY

As noted earlier, when deciding whether to bypass or reconstruct drainage of an obstructed kidney rather than excise it, the potential for meaningful recovery of function in the affected kidney represents a critical issue. In many cases, obstruction may be partial, so that it is difficult on the basis of the history alone to predict the outcome. In addition, imaging studies that reveal both anatomy and function of the obstructed kidney predict the extent of functional recovery poorly (see earlier discussion), because the extent of anatomic distortion during obstruction correlates poorly with the extent of recovery once the obstruction is relieved.³³² Isotopic renography with a variety of isotopes can be used to examine renal function, as outlined earlier. This approach is a far more reliable indicator of potential renal function when applied well after temporary drainage of the obstructed kidney (e.g., by nephrostomy tubes) has been achieved than if it is performed while the obstruction is still present.³³² Imaging

of the anatomy will provide information on the size and volume of the kidney but does not demonstrate reliable information on kidney function. All of these considerations figure into the clinical judgment as to whether attempts should be made to salvage the kidney. However, there are presently no methods available to predict reliably the functional potential recovery of an obstructed kidney.

In cases of prenatal urinary tract obstruction, clinical decision-making is complex because the risks of not intervening can be very high, as can the risks of prenatal surgery. Because fetal intervention can be associated with frequent complications and a high rate of fetal loss, subjects for intervention should be carefully chosen. Fetal renal biopsy, which demonstrated a 50% to 60% success rate, correlates well with outcome and has few maternal complications.^{10,12,13,310,318} It may be used as one of the methods to determine treatment strategy. Studies demonstrate that antenatal intervention may help fetuses with the most severe forms of obstructive uropathy, otherwise usually associated with a fatal neonatal course.^{10,12,13,316}

RECOVERY OF RENAL FUNCTION AFTER PROLONGED OBSTRUCTION

In patients the potential for renal recovery depends primarily on the extent and duration of the obstruction. However, other factors, such as the presence of other illnesses and the presence or absence of urinary tract infection, play an important role as well. In dogs subjected to 40 days of ureteral ligation, release of the obstruction led to no recovery of renal function. However, recovery of renal function in humans has been documented following release of obstruction of 69 days or longer.^{333,334} Because it is difficult to predict whether renal function will recover when temporary relief of obstruction has been achieved, it makes sense to measure function repeatedly with isotopic renography over time, before deciding on a definitive surgical course. Chronic bilateral obstruction, as seen in benign prostatic hyperplasia, can cause chronic renal failure, especially when the obstruction is of prolonged duration and when it is accompanied by urinary tract infections.^{334,335} Progressive loss of renal function can be slowed or halted by relieving the obstruction and treating the infection.

When obstruction has been relieved and there is poor return of renal function, interstitial fibrosis and inflammation may have supervened. To ensure that there is no other process hampering recovery of renal function, renal biopsy may be indicated. As noted earlier, studies in experimental animals have implicated a variety of factors in chronic renal failure due to prolonged obstruction, including excessive production of renal vasoconstrictors such as renin and angiotensin and growth factors that may enhance fibrosis. Based on these findings, inhibitors like captopril,³²² angiotensin receptor antagonists,³¹² NO supplementation,³⁰³ α -MSH,³⁰⁵ and EPO³⁰⁶ have been shown experimentally to ameliorate to some degree the long-term damage to kidneys observed following prolonged obstruction.

POSTOBSTRUCTIVE DIURESIS

Release of obstruction can lead to marked natriuresis and diuresis with the wasting of potassium, phosphate, and divalent

cations. It is notable that clinically significant postobstructive diuresis usually occurs only in the setting of prior bilateral obstruction, or unilateral obstruction of a solitary functioning kidney. The mechanisms involved have been described in detail earlier and involve the combination of intrinsic damage to tubular sodium, solute, and water reabsorption, as well as the effects of volume expansion, solute (e.g., urea) accumulation, and attendant increases in natriuretic substances such as ANP. When the obstruction is unilateral and there is a functioning contralateral kidney, the volume expansion, solute accumulation, and increases in natriuretic substances do not occur, and the contralateral kidney may retain salt and water, resulting in some compensation for the natriuresis and diuresis occurring in the postobstructive kidney. Management of the patient with postobstructive diuresis focuses on avoiding severe volume depletion due to salt wasting, and other electrolyte imbalances, such as hypokalemia, hyponatremia, hypernatremia, and hypomagnesemia.

Postobstructive diuresis is usually self-limited. It usually lasts for several days to a week, but may, in rare cases, persist for months. Acute massive polyuria or prolonged postobstructive diuresis may deplete the patient of Na^+ , K^+ , Cl^- , HCO_3^- , and water, as well as divalent cations and phosphate. Volume or free water replacement is appropriate only when the salt and water losses result in volume depletion or a disturbance of osmolality. In many cases, excessive volume or fluid replacement prolongs the diuresis and natriuresis. Because the initial urine is isosthenuric, with an initial Na^+ of approximately 80 mEq/L, an appropriate starting fluid for replacement may be 0.45% saline, given at a rate somewhat slower than that of the urine output. During this period, meticulous monitoring of vital signs, volume status, urine output, serum and urine chemistry, and osmolality is imperative. This will determine the need for ongoing replacement of sodium, free water, and other electrolytes. With massive diuresis, these measurements will need to be repeated frequently, up to four times daily, with frequent adjustment of replacement fluids, as needed. With relief of uncomplicated obstruction, the kidney function usually returns to normal with adequate hormonal responses. However, some cases, especially in infants with bilateral obstructive nephropathy, have been described where the distal tubule remains refractory to aldosterone (pseudohypoaldosteronism) and the patients develop paradoxical hyperkalemia despite a returning kidney function and a falling plasma creatinine.³³⁶ In such cases, during postobstructive diuresis, plasma potassium must be monitored carefully.



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BOARD REVIEW QUESTIONS

1. How is hydronephrosis defined?
 - a. An obstructed kidney with impaired drainage of urine
 - b. A kidney with an enlarged renal pelvis
 - c. An enlarged kidney
 - d. A kidney with reduced size and parenchymal scars
 - e. An enlarged kidney with the presence of kidney stones

Answer: b

Rationale: It is consistent with the anatomical changes.

2. How is urinary tract obstruction defined?
 - a. A condition with impaired kidney function
 - b. A condition with impaired kidney function due to vesicoureteral reflux
 - c. A condition where the surgeon finds the kidney obstructed
 - d. A condition where impaired urine drainage progressively leads to reduced kidney function
 - e. A condition that on computed tomography shows a hydronephrotic kidney

Answer: d

Rationale: It is consistent with the functional changes in kidney function.

3. Which imaging studies are appropriate for diagnosis and investigation of hydronephrosis in an adult patient?
 - a. Ultrasound for detection of hydronephrosis
 - b. Computed tomography to identify a cause for obstruction
 - c. Renography to assess differential renal function
 - d. Diuretic renography to distinguish obstruction from stasis
 - e. All of the above

Answer: e

Rationale: It is consistent with the added value of complementary examinations.

4. Why is renal blood flow and glomerular filtration rate impaired in response to urinary tract obstruction?
 - a. Because the increased pelvic pressure in the enlarged obstructed renal pelvis mechanically compresses the renal artery
 - b. Because urine drainage is reduced and the obstructed kidney does not require a normal blood supply
 - c. Because glomerular filtration pressure is increased in the obstructed kidney due to constriction of the efferent glomerular arteriole
 - d. Because obstruction leads to vasoconstriction of the afferent glomerular arteriole
 - e. Because obstruction leads to vasodilation of the afferent glomerular arteriole

Answer: d

Rationale: It is consistent with the well-characterized experimental evidence.

5. What are the pathophysiological mechanism/mechanisms responsible for obstruction-induced tubulointerstitial fibrosis?
 - a. Release by tubule cells of multiple proinflammatory and profibrotic molecules including tumor necrosis factor-alpha (TNF- α), transforming growth factor-beta (TGF- β), cytokines, and angiotensin II
 - b. Mast cell activation
 - c. Infiltration of the renal interstitium by inflammatory cells
 - d. Accumulation of interstitial myofibroblasts
 - e. All of the above

Answer: e

Rationale: It is consistent with the complexity of the condition.